Deutsche Bank
Research

CRO in 2Q26

Green, Green, Green

July 2026, Justin Bowers, CFA
Justin.Bowers@db.com
+1.917.517.9836 (m)

We continue to expect QoQ improvement over the course of 2026

— **2Q26 Earnings Outlook:** CRO stock performance entering 2Q26 is a much different story with the stocks +17% on average over the last month versus the group down (23%) on year-to-date basis entering the May earnings season. We remain bullish on CRO stocks and believe there is room to run from both positive estimate revisions into 2027 and recoupment of sector multiples compressed from AI / ML fears and lack of conviction in the cyclical recovery

— **1H26 Recap:** We reviewed our 2026 CRO Outlook from December to keep ourselves honest and when we were: most constructive on Charles River (CRL: Buy) due to a variant view; and, called Iqvia (IQV: Buy) is easier to own in 1H26 and Icon's (Buy: ICLR) 2026 guidance as likely to serve as a clearing event. CRL is the best performing CRO at +17% ytd and IQV and ICLR both down (8%) but setting up as positive estimate revision stories for 2027 with valuation support.

— **Buy Side Expectations:** The CROs are still under-owned although the Buy Side is above Sell-Side 2Q26 Consensus Book-to-Bills by 0.05x on average with ICLR at 1.30x the high watermark and MEDP bulls looking for 1.05x at the low-end. The stock performance in 4Q25 was painful the last time the Buy Side was over its skis but the sector has had a few quarters of Biotech funding accelerating and post-MFN deals for an acceleration in trial spend. Icon and Charles River remain favored CROs while sentiment is most negative for Iqvia (also screening as a consensus Long and Short in TMAC's most recent survey)

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

Macro indicators pointing up; we like CRL, ICLR, and IQV

— **AI as a Tailwind or Headwind 2.0:** We believe there is still multiple recovery from AI / ML fears alone amounting to 1-2x and our rationale for this only increases as we continue to channel work with large pharma and other industry participants. Our most recent check indicated a Top 5 Biopharma plans to maintain CRO outsourcing allocations longer term; is harvesting quality and speed-to-market from CRO AI / ML efforts without sacrificing cost; and, called out Iqvia as being the most progressive with tech deployment and business model changes: hence our recent Catalyst Call

— **Focus Stocks:** In CRO, we like **IQV** for upside to sentiment and an EPS beat and are also looking for beats and potentially modest positive revisions from **CRL** and **ICLR**

— **Macro Indicators: Green, Green, Green:** 2Q26 was the third greatest quarter on record for biotech equity funding; drug approvals tracking >2025; 2Q26 licensing levels +70% YoY; large pharma M&A on-track to exceed 2025 levels; increased preclinical / early-stage outsourcing to China / India (+5 – 10% vs. 2-3 years ago)

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

Updated PT for MEDP; CRL; and FTRE

— We are increasing our PT for MEDP to **$480 (from $425) based on a 28x 2026E EPS multiple (from 25x 2026E EPS)**; our target multiple is in-line with the 5YA forward multiple and reflective of recovering sector multiples / growth

— We are increasing our PT for CRL **to $260 (from $230) based on a 13.5x 2027E EBITDA multiple (from 12.5x 2027E EBITDA)**; the target multiple is in line with the stock's 5-year historical average due to improving end-market conditions and sector multiples

— We are increasing our PT for FTRE to **$18 (from $14) based of a 11x 2027E EBITDA multiple (from 8x 2027E EBITDA)**; our target multiple is a discount to the CRO peer group median, with greater margin expansion and EBITDA growth prospects offset by greater financial risk from elevated leverage and revenue growth headwinds this year.

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

DB Estimate Changes
Tweaking estimates to reflect intraquarter changes / 2027 estimates

		DB Estimates				vs Prior Estimates				+ / - Earnings	
	Mkt Cap	Revenue		Earnings		Revenue		Earnings		Related	
Ticker	($mm)	2026	2027	2026	2027	2026	2027	2026	2027	Implied Move	Notes
IQV	34,877	$17,158	$18,045	$12.62	$14.00	(0.3%)	(0.5%)	(1.8%)	(2.5%)	5.9%	Updated to reflect debt issued in 2Q26 and fx; no change to PT or multiple
MEDP	15,294	$2,794	$2,880	$17.05	$19.19	0.0%	0.0%	0.1%	0.6%	12.4%	2026 / 2027 EPS <+1.0%, updated book-to-bill; PT to $480 (from $425), multiple to 28x 2026E EPS (from 25x 2026E EPS)
ICLR	12,928	$7,996	$8,220	$10.46	$11.66	0.0%	0.0%	0.0%	0.0%	8.7%	No changes
CRL	11,252	$3,846	$3,889	$11.00	$12.50	0.0%	0.6%	0.0%	2.4%	9.4%	2027 EBITDA / EPS +2%; PT to $260 (from $230), based on a 13.5x 2027E EBITDA (from 12.5x 2027E EBITDA)
CON	4,018	$2,329	$2,465	$471.50	$507.22	0.0%	0.0%	0.0%	0.0%	2.2%	No changes
FTRE	1,672	$2,609	$2,668	$206.22	$219.51	0.0%	0.0%	0.0%	0.0%	13.4%	PT to $18 (from $14), multiple to 11x 2027E EBITDA (from 8x 2027E EBITDA)
MRVI	1,541	$211	$228	$30	$42	0.0%	0.0%	0.0%	0.0%	5.5%	No changes

Note: Data as of 7/10/2026; FTRE, SEM, CON, and MRVI earnings reflect EBITDA ($mm)
Source: Deutsche Bank, Bloomberg Finance LP, Company Reports

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

DB vs Consensus
Street generally more aggressive on 2Q26 book-to-bill and bookings

Ticker	Mkt Cap ($mm)	Consensus 1Q26	Consensus 2Q26	QoQ	DB 2Q26	DB vs Street	Buyside
Book-to-Bill							
IQV	34,877	1.04x	1.11x	0.07x	1.10x	(0.01x)	1.10 - 1.15x
MEDP	15,294	0.88x	0.98x	0.10x	0.95x	(0.03x)	1.05x
ICLR	12,928	1.42x	1.28x	(0.14x)	1.25x	(0.03x)	1.30x
CRL	11,252	1.04x	1.06x	0.02x	1.00x	(0.06x)	1.05 - 1.10x
FTRE	1,672	1.15x	1.06x	(0.09x)	1.10x	0.04x	1.15x
Bookings							
IQV	34,877	$2,500	$2,675	7.0%	$2,720	1.7%	
MEDP	15,294	$618	$658	6.4%	$665	1.1%	
ICLR	12,928	$2,880	$2,563	(11.0%)	$2,493	(2.7%)	
CRL	11,252	$622	$625	0.4%	$585	(6.4%)	
FTRE	1,672	$732	$686	(6.3%)	$711	3.7%	

Note: Data as of 7/10/2026
Source: Deutsche Bank, Bloomberg Finance LP, Company Reports

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

DB vs Consensus

More cautious than the Street on 2027 top line, relatively in-line on 2026 revenue

		Consensus		Consensus		DB vs Consensus			
	Mkt Cap	Revenue		Earnings		Revenue		Earnings	
Ticker	($mm)	2026	2027	2026	2027	2026	2027	2026	2027
IQV	34,877	$17,286	$18,276	$12.81	$14.17	(0.7%)	(1.3%)	(1.5%)	(1.2%)
MEDP	15,294	$2,779	$2,958	$17.02	$18.47	0.5%	(2.6%)	0.2%	3.9%
ICLR	12,928	$8,062	$8,310	$10.64	$11.86	(0.8%)	(1.1%)	(1.7%)	(1.6%)
CRL	11,252	$3,848	$3,923	$11.12	$12.32	(0.0%)	(0.9%)	(1.0%)	1.4%
CON	4,018	$2,334	$2,471	$471.88	$502.67	(0.2%)	(0.2%)	(0.1%)	0.9%
FTRE	1,672	$2,607	$2,689	$210.00	$234.83	0.1%	(0.8%)	(1.8%)	(6.5%)
MRVI	1,541	$211	$226	$31	$39	(0.0%)	0.8%	(3.9%)	7.7%

Note: Data as of 07/10/2026; FTRE, SEM, CON, and MRVI earnings reflect EBITDA ($mm)
Source: Deutsche Bank, Bloomberg Finance LP, Company Reports

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

Durability of the recovery; 4Q26 into 2027; EBP; pricing; and margin headwinds

— **Durability / Momentum of Improvements:** 1Q26 CRO results / commentary were broadly supportive of improved demand; debate is around whether it represents just further stabilization or a real inflection

— **4Q26 Exit Rates:** Investors are sizing the range of QoQ improvements before hitting 4Q26 and whether 2027 will represent a return to more LRP-type CRO growth

— **Continued EBP Improvement:** Indications are that large / mid-sized pharma remain stable; EBP RFPs and bookings improving but multiple quarters of improvement likely to be required as proof

— **FSO / FSP and Pricing:** FSP penetration still strong with large pharma, more FSO to enter the mix as EBP returns; pricing stabilizing but still hasn't flipped back to CRO's favor

— **Margin Headwinds:** Mix / pricing / quality of bookings all driving margin headwinds in 2026; higher quality bookings achieved in late-2025, with the potential to positively impact margins in 4Q26

— **Trial Concentration:** Cardiometabolic studies remain in-vogue; uncertainty around whether over-indexing to certain trial types could flip to a headwind

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

Biotech / preclinical observations; company – specific observations from Field Work

- **Biotech:** Optimism remains elevated across sector while RFP TaT and decision making improves; cycle time shortening from ~9 months to sub-6 months referenced by a private CRO
- **RFP Flow / Awards Improvement:** Private CRO indicated 1H26 RFP flow +33% YoY with awards improving YoY as well; both metrics also improved QoQ in 2Q26, VC capital flowing to biotech companies much easier
- **Higher-Quality RFPs Available:** 2nd private CRO noted total RFP $'s were stable YoY, but quality of RFPs significantly improved with better funding / quality of data and less ballparks
- **CRO Bid Process Narrowing:** Less CROs invited to bid process, now 3 – 4 similar-sized CROs vs. 7 – 8 last year; large pharma still inviting larger numbers of CRO to bid
- **Pricing:** Pricing environment is stabilizing after a couple years of price pressure; expect flattish to low-SD price for functional service provider (FSP) and full-service outsourcing (FSO); CRO's force-ranked from most to least expensive as: Iqvia; Fortrea; Icon = Parexel; PPD; and Medpace
- **China / India Commanding More Early-Stage Work:** +5 – 10% more preclinical / early-stage work in China / India vs. 2 – 3 years prior at (35 – 20%) cheaper / +40% faster; expectations for volume ex-U.S. to increase over time
- **Biotech CRO Share Changes:** Icon taking share from Fortrea; Fortrea among the more expensive biotech-focused CROs

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

Broad use among large pharma, benefits still unclear

— **Top 10 Biopharma:** Referenced using AI for 7 years including a home-based program to draft reports. Is seeing savings in translation and using the savings to spend more with CRO partner

— **Top 10 Biopharma:** Potential savings could eventually deliver a (20%) reduction in clinical monitoring costs, maybe a (10%) reduction in CRAs. Plans to reinvest in more programs vs . reduce R&D spend.

— **Top 10 Biopharma:** Still anticipates the work needs to get done manually, leading to no layoffs; more focused on efficiency

— **Top 10 Biopharma:** AI used in early stage and discovery; think there's a scenario to use AI and go from 10mm to 20mm targets with less manual screening

— **Top 10 Biopharma:** Sees opportunities for savings in areas like biostatistics, documentation, clinical monitoring - maybe 10% savings. AI is likely be to impactful over the next 3-5 years, will reinvest savings

— **Midsized Biopharma:** There is a cost for the CRO to validate and implement the technology so it's important not to nickel and dime the sites and CROs. Still five years out from cost savings and it requires investment

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

Leveraged across CROs to drive efficiencies, human element still necessary

— **Precommercial Biopharma:** Will not turn away a vendor or a CRO if it has it, but lacks the scale or the number of clinical programs to justify the investment in AI.

— **Industry Trade Group:** Thinks AI will augment, not replace clinical monitoring

— **Private CRO:** Transparency on AI-use is important in relationship with sponsors; pricing rates are the same but there will be less time utilized and there is room for budget efficiency with AI

— **Private CRO:** Clients haven't changed RFP processes due to AI; its still a 10-day response. What has changed is that its now 10 CROs versus 3 CROs at the table

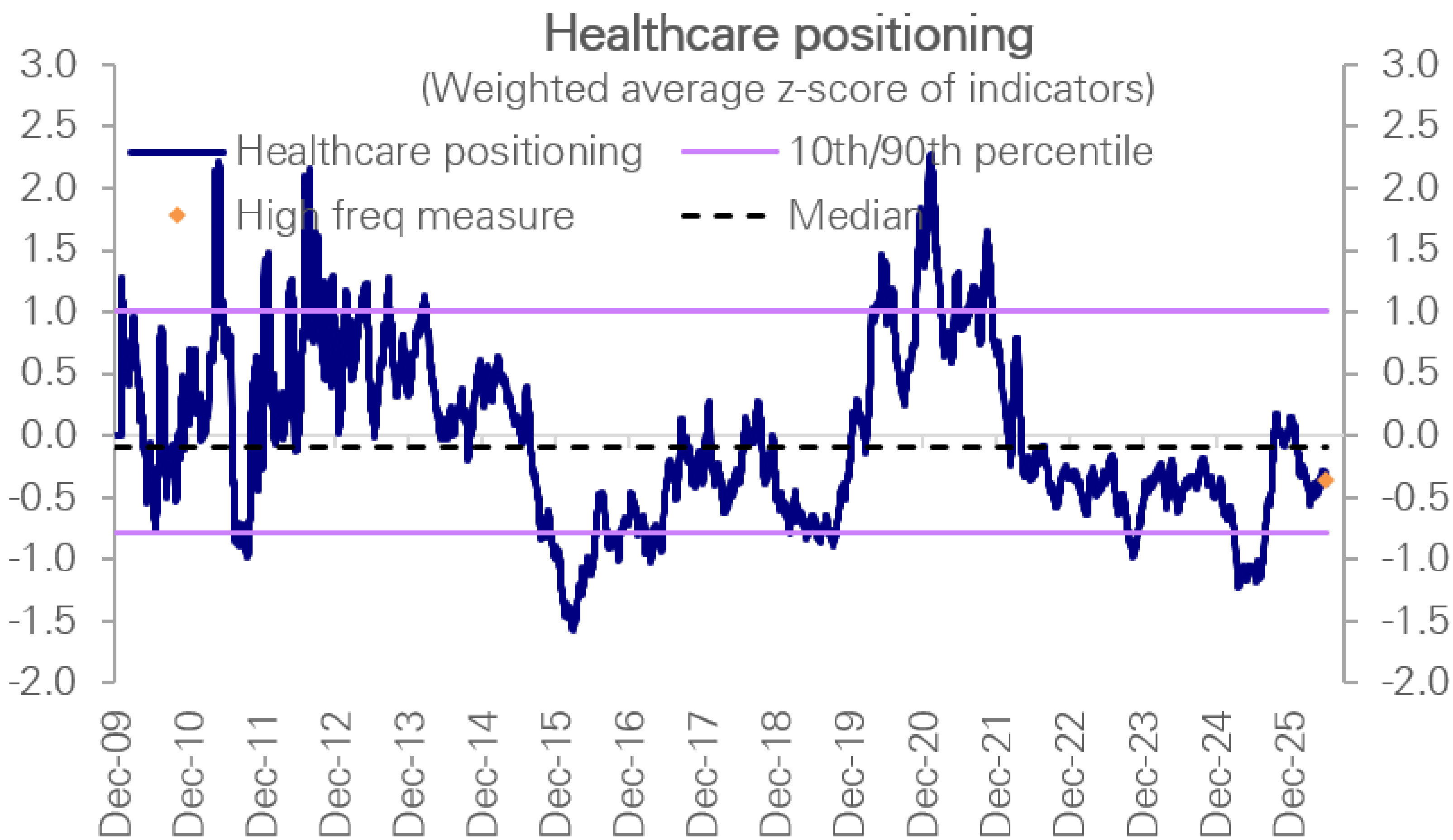
— **Private CRO:** AI / ML will be margin accretive to the company in the interim, will have to give back some of the savings to sponsors over time

— **Private CRO:** Sector margins will be pressured over the next few years due to pricing, the industry might be able to maintain margins with offsets from AI and productivity

— **Large Pharma Outsourcing Specialist:** AI unlikely to fully remove the human element from tasks it replaces; protocol writing; medical writing; statistical analysis; data management, review, and quality; and systems programming were all mentioned as at-risk

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

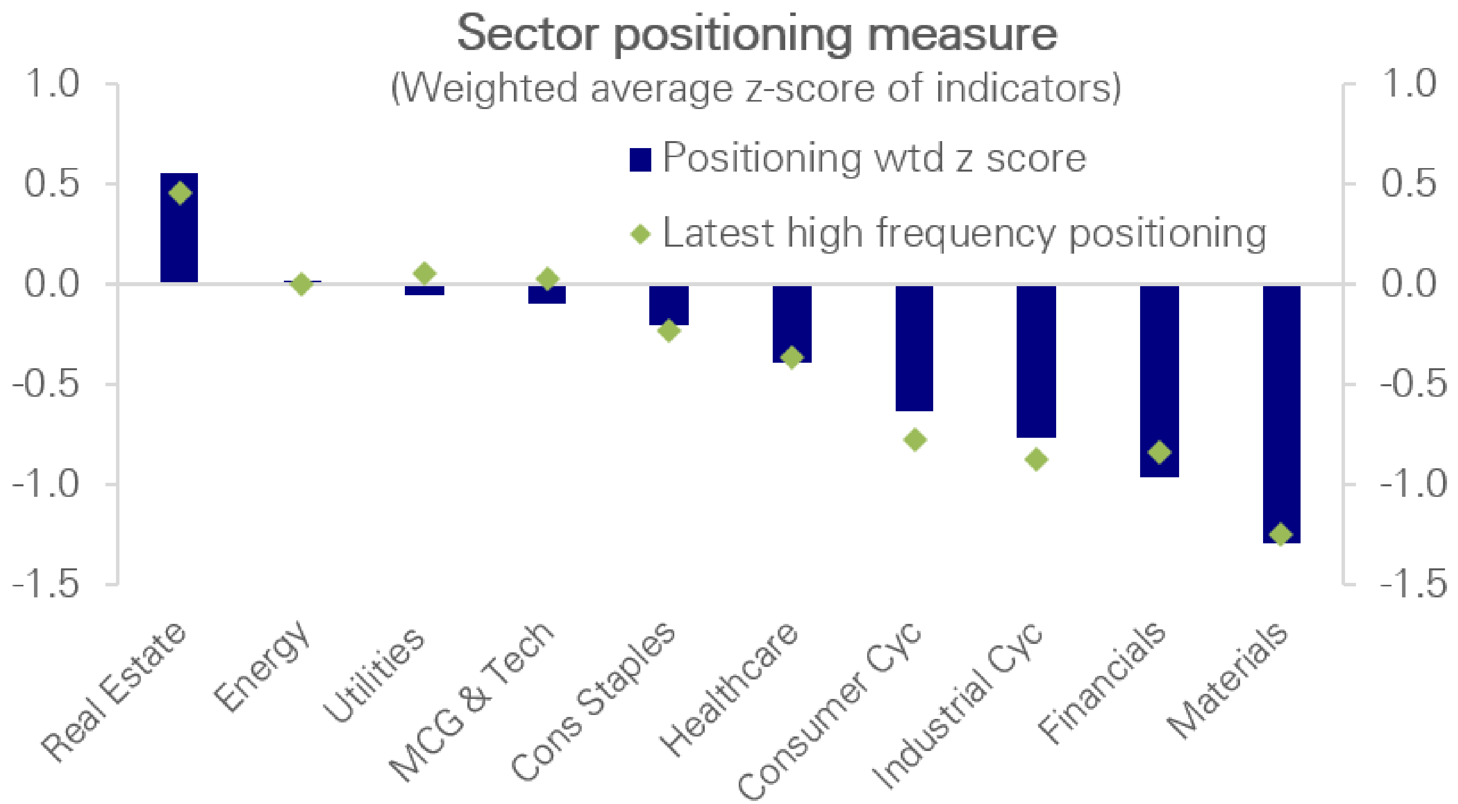
Healthcare at 33%ile; below median



Note: Most recent percentile of 33%ile as of 7/7/26
Source: Deutsche Bank Asset Allocation

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

Equity positioning across sectors



MCG is Mega-cap Growth; Financials excludes MA & V which are under MCG; Industrial cyc: Industrials ex pandemic-hit cos; Cons cyc: Cons Disc & Media ex MCG ex Restaurants ex pandemic-hit cos.

Source: Deutsche Bank Asset Allocation

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

Comp Sheet

Multiples re-rating upward over the course of 2Q26 across Tools / Pharma Services

($ Millions, except p/s and multiples)			Price		% to	Mkt Cap	EV	CY SALES*			CY EBITDA*			CY EPS*		
Company	Ticker	Rating	7/9/26	Price Tgt	Price Tgt	(US$ mm)	(US$ mm)	2025	2026e	2027e	2025	2026e	2027e	2025	2026e	2027e
AVANTOR	AVTR	Hold	10.51	11.0	5%	7,094	10,631	6,552	6,501	6,635	1,181	972	1,015	1.02	0.79	0.85
CHARLES RIVER	CRL	Buy	233.60	260	11%	11,243	14,155	4,015	3,848	3,923	977	993	1,055	10.29	11.12	12.32
DANAHER	DHR	Buy	195.98	250	28%	140,882	154,873	24,568	25,596	27,217	7,689	8,194	8,886	7.80	8.45	9.13
FORTREA	FTRE	Hold	17.67	18	2%	1,680	2,651	2,723	2,607	2,689	190	210	235	0.43	0.88	1.08
ICON PLC	ICLR	Buy	168.77	188	11%	12,848	15,590	8,251	8,062	8,310	1,531	1,330	1,422	12.52	10.64	11.86
IQVIA	IQV	Buy	208.97	240	15%	34,690	48,780	16,310	17,287	18,278	3,785	3,999	4,251	11.79	12.81	14.17
MARAVAI	MRVI	Buy	5.97	6.00	1%	1,572	1,677	186	211	226	(35)	31	39	(0.30)	(0.11)	(0.09)
MEDPACE	MEDP	Hold	535.50	480	(10%)	15,372	14,865	2,530	2,779	2,958	470	611	660	12.81	17.02	18.47
REPLIGEN	RGEN	Buy	143.94	180	25%	8,136	8,039	738	823	942	142	167	207	1.71	2.00	2.54
BIO-TECHNE	TECH	Hold	71.15	73	3%	11,163	11,244	1,219	1,224	1,299	430	437	473	1.92	1.98	2.16
THERMO FISHER	TMO	Buy	524.71	630	20%	195,863	235,896	44,556	47,754	50,319	10,770	12,300	13,289	22.87	24.87	27.22
10X GENOMICS	TXG	Hold	43.10	40	(7%)	5,425	4,966	643	614	668	(114)	(73)	(57)	(0.99)	(0.76)	(0.67)
WATERS	WAT	Hold	377.13	380	1%	36,960	42,068	3,165	6,440	7,104	1,122	2,170	2,425	13.13	14.51	16.50
WEST	WST	Buy	357.66	405	13%	24,989	24,784	3,074	3,333	3,536	781	918	1,011	7.29	8.61	9.58
CONCENTRA	CON	Buy	31.40	32	2%	4,037	6,127	2,163	2,334	2,471	432	472	503	1.36	1.51	1.69

		YTD	D'ly lqid	Short Int	Short Int	PEG Ratio	EV/SALES*			EV/EBITDA*			P/E*		
Company	Ticker	Price Chg	('000s)	(days)	Float (%)	2 Yr EPS CAGR	2025	2026e	2027e	2025	2026e	2027e	2025	2026e	2027e
AVANTOR	AVTR	(9%)	83,835	6	8%	nm	1.6x	1.6x	1.6x	9.0x	10.9x	10.5x	10.3x	13.2x	12.2x
CHARLES RIVER	CRL	17%	160,097	4	9%	2.2	3.5x	3.7x	3.6x	14.5x	13.9x	13.0x	22.7x	21.0x	18.9x
DANAHER	DHR	(13%)	840,204	2	2%	2.9	6.3x	6.1x	5.7x	20.1x	18.8x	17.3x	25.1x	23.6x	21.8x
FORTREA	FTRE	3%	20,641	5	10%	0.3	1.0x	1.0x	1.0x	14.0x	12.3x	11.0x	40.8x	20.1x	16.4x
ICON PLC	ICLR	(8%)	156,977	2	4%	nm	1.9x	1.9x	1.9x	10.2x	11.6x	10.9x	13.5x	15.8x	14.1x
IQVIA	IQV	(8%)	290,988	3	3%	1.7	3.0x	2.8x	2.7x	12.9x	12.1x	11.4x	17.7x	16.2x	14.7x
MARAVAI	MRVI	87%	13,387	2	8%	nm	9.0x	8.0x	7.4x	nm	nm	nm	nm	nm	nm
MEDPACE	MEDP	(4%)	171,471	6	9%	1.6	5.9x	5.3x	5.0x	31.7x	24.1x	22.3x	41.8x	31.6x	29.1x
REPLIGEN	RGEN	(12%)	141,867	5	11%	3.3	10.9x	9.8x	8.5x	nm	47.3x	38.1x	nm	72.1x	56.8x
BIO-TECHNE	TECH	21%	280,329	1	10%	6.1	9.2x	9.3x	8.9x	26.2x	26.0x	24.6x	37.1x	37.1x	34.7x
THERMO FISHER	TMO	(9%)	1,117,304	2	1%	2.3	5.3x	4.9x	4.7x	21.9x	19.2x	17.8x	22.9x	21.2x	19.4x
10X GENOMICS	TXG	162%	87,852	4	14%	nm	7.7x	8.1x	7.4x	nm	nm	nm	nm	nm	nm
WATERS	WAT	(1%)	356,381	3	4%	2.1	13.3x	6.5x	5.9x	nm	19.2x	17.2x	28.7x	25.9x	22.8x
WEST	WST	29%	267,568	2	3%	2.8	8.1x	7.4x	7.0x	31.7x	26.9x	24.4x	49.1x	41.1x	36.9x
CONCENTRA	CON	60%	23,890	3	2%	1.8	2.8x	2.6x	2.5x	14.2x	11.8x	11.1x	23.1x	20.8x	18.7x
Median		**(1%)**		**3**	**8%**	**2.2**	**5.9x**	**5.3x**	**5.0x**	**14.5x**	**18.8x**	**17.2x**	**24.1x**	**21.2x**	**19.4x**

Note: Data as of 7/9/2026
Source: Deutsche Bank, Bloomberg Finance LP, Company Report

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

Field Work (Jan 2026 to Jun 2026)
Primary Research via industry contacts, customer feedback and site visits

Date	Title	Highlights
7/7/2026	Field Notes: Large Biopharma R&D Outsourcing Perspectives in July 2026	We had a broad discussion on large biopharma R&D outsourcing trends and developments with a specialist involved in vendor selection and with over 17 years of experience. Large pharma outsourcing remains consistent with our conversation from ~1 year ago; FSP penetration at 70 – 75% of large pharma work was deemed “stable” with trends towards outsourcing in: a) specific services; and b) investigational / early-stage work to cheaper regions (China / India).
6/8/2026	Field Notes: U.S. A&G Bottom is in; Numbers Inflecting to Upside Risk	Our biggest takeaways from recent incremental diligence - informed primarily by channel work around the American Society of Mass Spectrometry meeting (ASMS preview here) - is that the bottom is in, in the U.S. Academic and Government (A&G) end market, and is improving QoQ.
6/3/2026	10x Genomics: Spatial at Scale; Highlights from the Atera World Tour, ASMS Nuggets	We recently attended 10x Genomics’ Atera World Tour, New York edition, and assessed the company’s newest addition to its instrument portfolio for the first time in-person. We also spoke with several attendees and users of 10x Genomics’ technology at that event and at the ASMS meeting this week.
4/28/2026	Field Notes: GLP-1 (++); Reshoring (+); Capex / A&G (Neutral); Intraquarter CY1Q26 EPS Updates	We attended Interphex last week and met with more than 30 Tools / biopharma-adjacent companies. Our key takeaways from our day in the field were: 1) reshoring is very real and in its early-innings, with a longer runway than we originally appreciated; 2) sentiment around early-stage / biotech was neutral to improving, in-line with company commentary from 4Q25 / so-far in 1Q26; 3) CapEx, while still challenged, has troughed and doesn’t seem poised for another step down; and 4) injectable GLP-1s remain a key driver for demand.
4/16/2026	Contract Research Organizations: Takeaways from Clinical CRO Specialist with Biotech Focus, April 2026	We hosted a CRO specialist with 30+ years of industry experience and who is currently an executive at a private CRO focused on early-stage clinical biotech. The specialist's commentary on demand indicators is directionally positive and consistent with recent intraquarter checks. The specialist cited +low-double-digit bookings (and RFP volume) growth for biotech in 1Q26. He also expects this trendline to continue in 2Q26 based on the current funnel and a very strong 4Q26. VC sponsors are showing an increasing appetite to take risk, deploy capital, and fund greater numbers of programs compared to prior years.
3/18/2026	Contract Research Organizations: Takeaways from Call with CRO / Pharma Vendor Specialist	We hosted a call with a CRO / pharma vendor specialist with 30+ years of industry experience across the broader CRO, CDMO, and pharma packaging landscape. The specialist’s views on pricing and AI diverge from prevailing sentiment with pricing power still in the hands of CROs for more specialized therapeutic areas and AI accretive to CRO industry growth over the interim. The specialist indicated that larger CROs are now charging for AI services in its RFPs and cited increased demand for Iqvia's RWE and data analytics-based tools.
3/25/2026	Charles River Laboratories: Updates from the Field, NAMs and Positive Estimate Revisions	Our discussions across the channel the last two days with sponsors, service providers, and regulatory bodies reinforce the prior industry view(s) that the vast limitations of alternative testing methods are still years if not decades away from being substitutes or even co-validated safety assessment methods for preclinical drug development. If we assume that the experts involved in the day-to-day of drug development are correct and prescient then: 1) the multiple contraction over the last two months in CRL stock has been unwarranted; and, 2) an opportunity exists because the market is inefficient on this issue.
3/18/2026	Contract Research Organizations: Thoughts and Industry Feedback on AI Contagion Part II: Industry Scenarios	We are providing updated thoughts on the impact of AI / ML on the CROs based on incremental diligence and recommend buying the stocks on the (31%) average multiple compression since January and driven by the market's concerns on terminal value due to AI / ML threats. We believe the market is pricing in low-single digit industry growth over the next cycle with forward PE multiples in the high-SD to low-DDs (our Diminishing R&D ROI Scenario).
1/22/2026	Contract Research Organizations: Takeaways from Clinical CRO Specialist with Biotech Focus, January 2026	We hosted a CRO specialist with 30+ years of industry experience and currently an executive at a private CRO (top 15 in industry) focused on early-stage clinical biotech. The discussion was directionally consistent albeit with a less optimistic forward outlook than recent January 2026 / December 2025 industry fieldwork highlighted in our JPM Healthcare Conference Debrief.

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

Field Work (Apr 2025 to Nov 2025)
Primary Research via industry contacts, customer feedback and site visits

Date	Title	Highlights
11/21/2025	Field Notes: Thoughts on 2026+ from a Clinical CRO Specialist with Biotech Focus	We had a wide ranging discussion with a CRO specialist with greater than 20 years of industry experience. The specialist sees a strengthening R&D investment cycle through 2029 (peak LOEs) while the pace of the recovery will be somewhat contingent on the interest rate cycle and how aggressively the Fed cuts rates in 2026.
10/10/2025	Field Notes: Preclinical Recovery on the Horizon; Monitoring Government Channel	Our discussions across the preclinical and early stage R&D channel point to tightening safety assessment capacity (DSA) with lead times beginning to extend beyond eight weeks and into 2026; The most variant data points we picked up would lead to a return to through-cycle market growth levels in 2026 (+mid-SD) if we extrapolate early budget indicators from two top 30 biopharmas (by R&D spend).
10/8/2025	Contract Research Organizations: Takeaways from Clinical CRO Specialist with Biotech Focus, October 2025	We hosted a CRO specialist with 30 years of industry experience and his observations of the current industry environment are mixed; Competition and pricing pressure remain elevated but consistent with 2Q levels. The government shutdown is escalating uncertainty in the U.S. following modestly improving sentiment related to FDA / HHS / NIH
9/12/2025	Life Science Tools and Diagnostics: 2025 DB Healthcare Summit Takes: Tools & Biopharma Services	We hosted number of companies across our coverage universe at the 2025 DB Healthcare Summit, notably at Icon, Iqvia, and West; CROs remain cautiously optimistic about the emerging trends the industry saw in 2Q25;We also discussed the Biopharma R&D Outsourcing Landscape with pharma and CRO industry thought leaders.
9/8/2025	Medpace: Highlights from Global Headquarters Site Visit	We met with Medpace management (President, CFO, IR, Associate Director of Clinical Laboratory) and toured the Clinical Labs at Cincinnati headquarters; The company noted that the 2H book-to-bill target of 1.15x is not formal guidance; The big question remains 2026E which we believe should show growth in 1H with management eyeing a 1.15x book-to-bill and the increasing mix of Cardiometabolic studies.
8/21/2025	Field Notes: Large Biopharma R&D Outsourcing Perspectives in 2025 / 2026	We had a wide-ranging discussion on the large biopharma R&D outsourcing industry landscape with a specialist involved in vendor selection; the specialist believes that the decline in industry NHP usage has largely plateaued, i.e., the rate of decline has decelerated if not abated after accelerating downward over the last five years.
7/3/2025	Contract Research Organizations: Takeaways from Clinical CRO Specialist with Biotech Focus	Biotech end-market QoQ is facing downward pressure on 2Q book-to-bills. Increase funding challenges; heightening competition with pricing pressure; less work to go around; and Macroeconomic uncertainty. Both revenue and bookings growth are trending low-double digit
6/25/2025	Field Notes: Biotech Desert	Biotech softening QoQ post subdued 1Q with extended decision making and price pressure; large pharma delaying awards with trade / MFN uncertainty; nearing end of reprioritization
6/5/2025	Field Notes: Subdued U.S. Lab Activity to Extend Into 2026	First instance of widespread cautiousness on 2026 spending; incrementally negative on U.S. A&G end market and broad capex spend over NTM
5/21/2025	Field Notes: 2Q Trending Better than Stocks Imply; Bioprocessing, CDMO Demand Intact	Resilient demand trends in Bioprocessing and CDMO midway through 2Q; acceleration of preclinical study starts; U.S. biopharma capex investment driving incremental inbound interest
4/9/2025	Contract Research Organizations: Takeaways from Clinical CRO Specialist with Biotech Focus	Pricing competitive but stable; greater competition for less available work (lower demand); HHS / FDA changes increasing caution, tariffs (+/-); ICLR cutting costs to protect margin; MEDP turnover / potential execution challenges
4/9/2025	Life Science Tools and Diagnostics: GLP-1 and Injectable Drug Handling Specialist Takeaways	Injectable Drug mfg. workflow; \$11.50-\$16.00 drug product unit cost for high throughput commercial runs and \$30.50-\$39.00 for clinical batches; device cost for autoinjector / pen injector is \$5.00-\$15.00; positive readthrough for WST Drug Handling
4/3/2025	Field Notes: Notes From The Floor: CDMO (+), Lean Into Bioprocessing	Conversations point to recovering biotech demand; stability in large biopharma; intensified bioprocessing competition; more positive outlook on CGT projects; CDMO activity stable QoQ from 2H and normalized lead times

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

Field Work (Mar 2024 to Mar 2025)
Primary Research via industry contacts, customer feedback and site visits

Date	Title	Highlights
3/20/2025	Life Science Tools and Diagnostics: CDMO Specialist Discussion: Increasing 1Q Activity, Business as Usual	Funding allocation shifted from 50% / 30% / 20% preclinical / clinical / follow on investment to 20% / 50% / 30%; biotech activity increasing; investor interest in GT improving; focus on China licensing deals
2/26/2025	Life Science Tools and Diagnostics: NIH Specialist Discussion: The Sky isn't Falling, but it is Raining	Exceptions for new project starts under CR; de minimis impact if NIH cut to large biopharma in next 3-5 years; greater clarity into NIH review process and do not expect material, dramatic declines in A&G end market
1/8/2025	Contract Research Organizations: CRO Specialist Discussion: Q4 In-Line with Q3; Improving Gradually in 2025	Robust Phase I activity; improving / stabilized pricing environment; anticipates longer recovery for biopharma demand; RFP volume strong, grew QoQ
12/12/2024	Field Notes: Research Triangle Round-Up; CRO Specialists See Ice Thawing	70-90% of large biopharma are through pipeline re-prioritizations, development model changes, and strategic partnerships/pricing. Biotech market accelerating; pockets of firming price
12/10/2024	Maravai LifeSciences: Flanders Site Visit: Build it and They Will Come	2024 is likely post-C19 trough and field work points towards firming biotech market and trough in preclinical market; Flanders 1 and 2 facilities demonstrate Maravai's technological leadership and differentiated value prop as an mRNA product and services provider
10/10/2024	Observations from a Top Gene Therapy CDMO Site Visit: 2024 Edition	Increased PD activity and GMP production; Improving activity from GT customers both RUO and GMP; TAM-expansion GT programs in-flight
10/10/2024	Preclinical Green Shoots; Biotech Customer Activity Growing	Waning pipeline rationalization; green shoots in biotech; stable but elevated NHP pricing; appropriate capacity; steady pricing above pre-C19
10/9/2024	Positive Biotech Demand Signal Across Drug Development Value Chain	Strengthening biotech end market; pick up in overall preclinical activity and increasing RUO material production
10/2/2024	CRO Specialist Discussion: Competition Elevated; Light at End of Tunnel	Strong Sept. exit for bookings with improving growth outlook through 2025 ; aggressive pricing in market; elongated decision making
9/19/2024	CRO Specialist Discussion: Structural Drivers Intact; Accelerating Growth	CRO market growth accelerating to 5-7% ; increasing competition; stable biotech demand; R&D dollars starting to flow to Phase I / preclinical
9/10/2024	Workers' Compensation Specialist Call: Ahead by a Country Mile (CON)	Customer portal, scale, and partnership drive market leadership; aging workforce; increased severity; healthy employment levels
8/20/2024	Bioprocessing KOL Discussion: Bioreactor, Resin Selection	Sartorius the easiest bioreactor to use; Danaher's moat in resins; increased biotech pressure
7/19/2024	CRO Specialist Discussion: RFPs Healthy but Extended; Competition	Positive-to-neutral views on Icon, Iqvia, and Fortrea; negative-to-neutral views on Parexel, Syneos, PPD, and Medpace; RFPs revisions up
6/26/2024	Upgrade to Buy Following Marlborough Site Visit (RGEN)	Base case IRR of 20-30% over 2-3 years; trough EBIT margins; increased depth of leadership; ATF / TTF leader
6/20/2024	Mixed Industry Datapoints; Clinical Trials Take the Cake	Increase in biopharma budgets for discretionary IT / Data; elevated clinical trial occupancy rates; AI / ML across large biopharma
5/22/2024	Highlights from Management Meetings; Near Term Optimism (WAT)	NDR with Waters management; Empower software; improving funnel in China; investors question instrument exposure and seasonality
4/19/2024	CRO Specialist Discussion: Increasing Bookings, Shifting Competition	Large CROs shift downstream; mid-sized CRO growth; reduced cancellations; labor market healthy
4/18/2024	CDMO and Bioprocessing Specialist Discussion; Market Firming	CDMO lead times +1 month in 2024; suspension > adherence; CTLT/NOVO impact
4/11/2024	CDMO KOL Discussion: Positive Signs for Cell & Gene Therapy; Pricing Healthy	Uptick in CGT demand; inbound inquiries from Biosecure risk; +2-5% price for CDMOs
3/22/2024	Shrewsbury Site Visit and SOT Debrief, Constructive on the Stock (CRL)	Increased pre-clinical activity in 2024; stable NHP price; 3-6 month booking timelines
3/21/2024	CDMO KOL Discussion: Early-Stage Recovery, Evolving Industry Dynamics	Headline risk from Biosecure Act; 12 month lag for improved biotech funding to materialize

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

AI-related fears overblown; margin bridge safe, potential upside to 2026 EPS

- IQV down (8%) YTD and is underperforming other CRO due to concerns about AI disintermediation; we believe these concerns are overstated and present an attractive entry point for a market leader with a proprietary data most in a heavily regulated industry
- We see modest upside to 2026 EPS due to improving end markets and capital deployment; investment in clinical development by large pharma and biotech is accelerating following the post-c19 reprioritizations, geopolitical uncertainty, and a resurgence in biotech funding, while Iqvia make inroads to the emerging biotech market.
- Recent industry checks indicate that large pharma is likely to outsource more due to AI; Iqvia is positioned as one of the biggest CRO AI-winners given its extensive suite of proprietary data / early-mover advantage with >190 AI-agents deployed (plans for =500 by YE27) and 19 / 20 top pharma using Iqvia agents today
- Only a minor piece of Iqvia's business (~$100mm) is at-risk from AI-related disruption and still grew +mid-single digit in 1Q26
- Concerns on the 2026 margin bridge ignore the underlying operational leverage of +60 bps of margin improvement that will carry through 2Q-4Q while fx and mix headwinds fade throughout the year
- IQV is trading at a current valuation at 11x 2027E EBITDA a 3x discount to the 5Y average at 13x, an attractive entry-point on a quality business.

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

Focus Stocks; IQV Cont.
IQV returns over the LTM

Note: As of 7/9/2026
Source: Bloomberg Finance LP

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

Preclinical activity returning; margin expansion baked in with 2H26 positive organic growth

— After exiting 1Q26 results with more conviction in our constructive thesis on CRL which was only strengthened following diligence from private competitors / upbeat meetings with the C-Suite

— Both divestments have been completed, and the company remains confident in >+500bps of 2H26 operating margin improvement vs. 1H26 with good line of sight

— The company also indicated there could be more wood to chop on efficiencies / costs, which could extend the margin expansion into 2027+

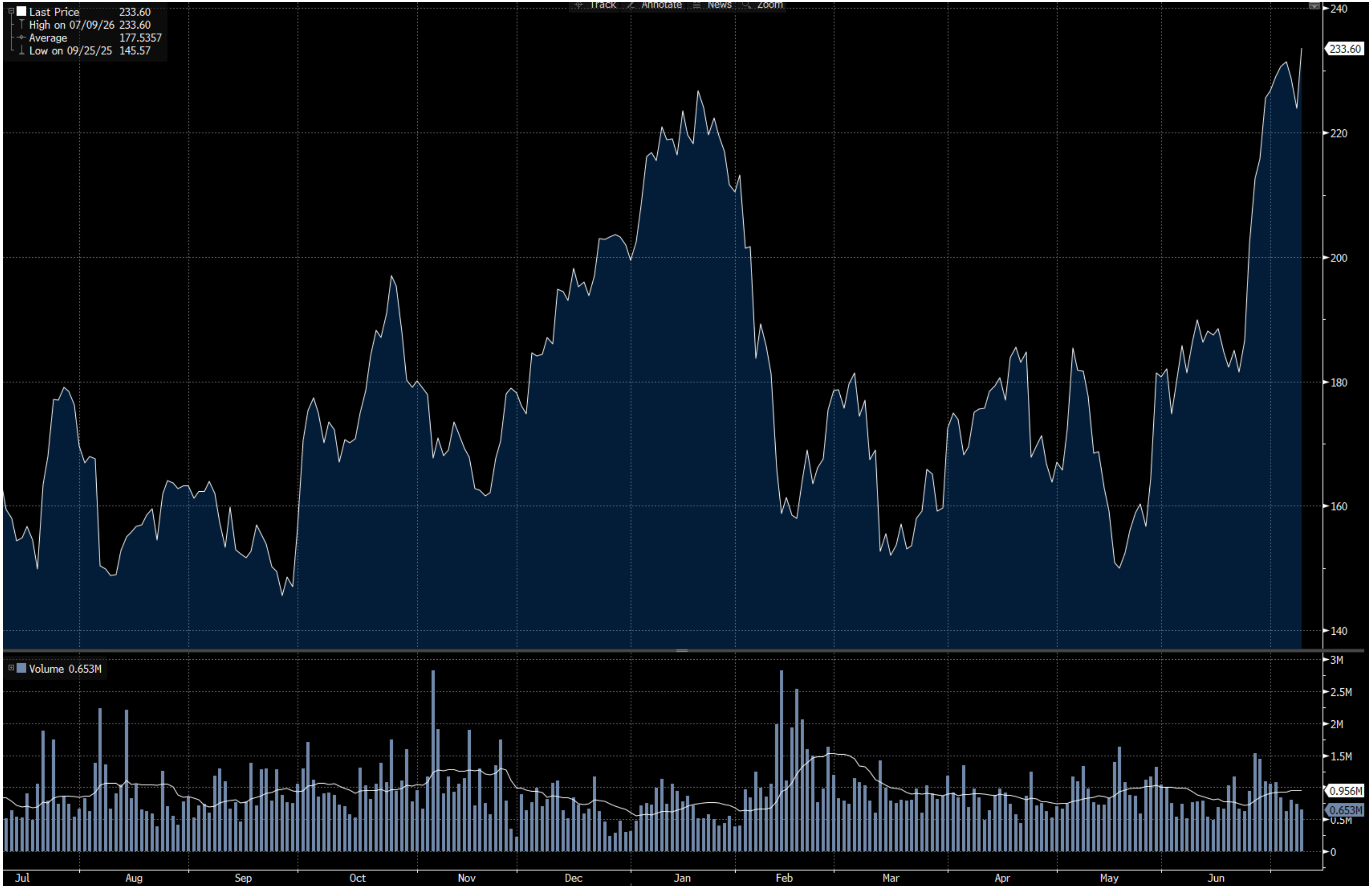
— 1Q26 demand metrics were supportive of a recovery from key customer cohorts and +mid-to-high-single-digit RFP growth is in-line with pre-C19 growth; Charlie anticipates further acceleration off +high-single-digit RFP growth

— Charles River still expects AI will be a tailwind overall to the broader CRO market and maintains a leading positions in NAMs (~$200mm in revenue)

— Current valuation at 13x 2027E EBITDA is in-line with the 5YA as investor confidence in the recovery has increased; we expect a multiple re-rating upward could be in-play should the company continue to execute and return to positive organic growth

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

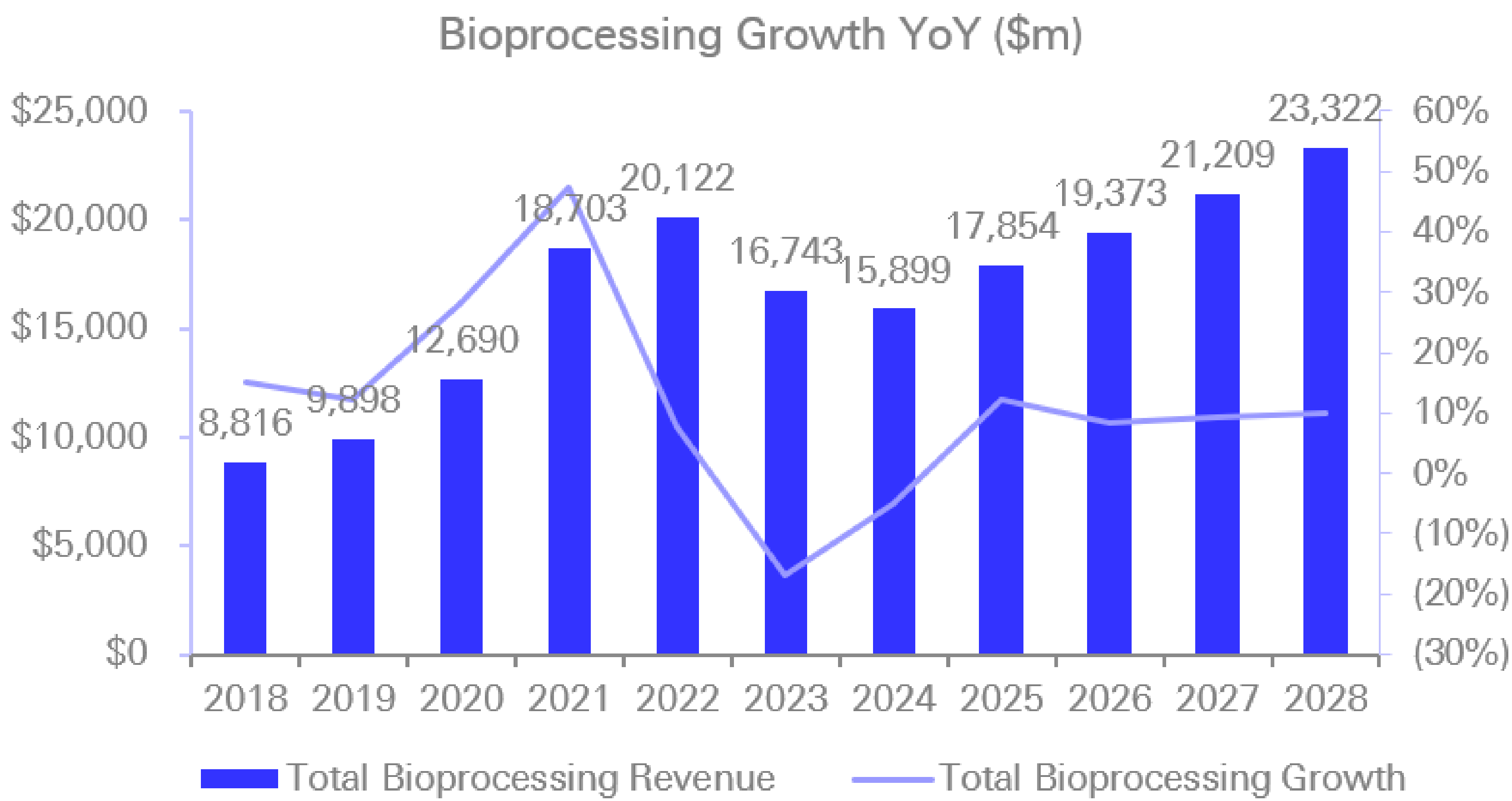
Focus Stocks; CRL Cont.
CRL returns over the LTM



Note: As of 7/9/2026
Source: Bloomberg Finance LP

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

Bioprocessing Industry Growth
High-single-digit growth in 2026



Note: The group includes Danaher, Thermo Fisher, Sartorius, Merck KGaA, and Repligen
Source: Deutsche Bank, Company Reports

Large Biopharma R&D Trends

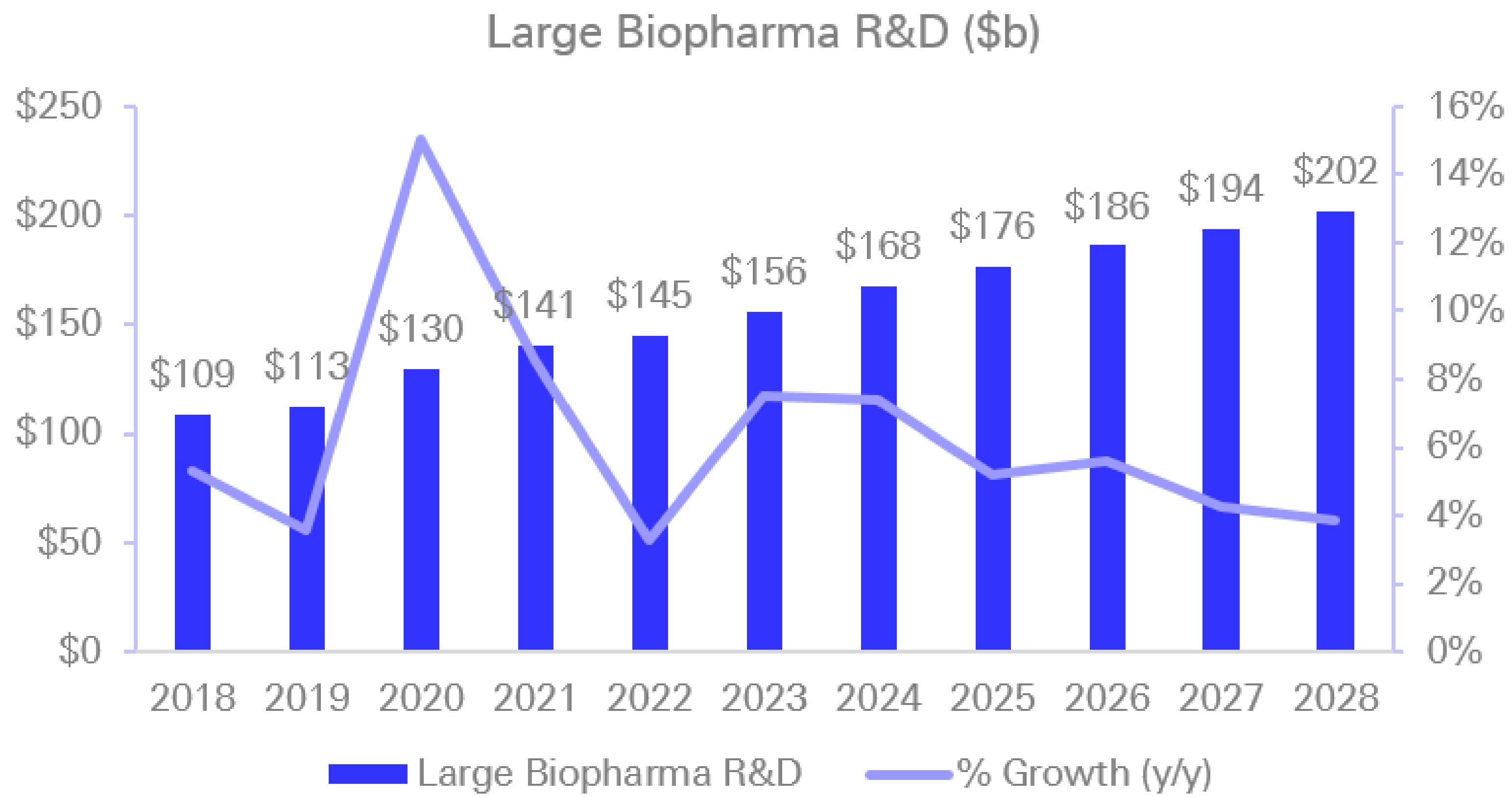
2026 R&D estimates flattish vs. April; +mid-single-digit R&D growth in 2026

Large Biopharma R&D Expenditures	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
ABBVIE	$5,830	$6,518	$6,435	$7,003	$7,711	$8,985	$9,756	$10,411	$10,965
ASTRAZENECA	$5,872	$7,987	$9,500	$10,165	$12,116	$13,822	$14,822	$15,234	$16,034
BAYER	$8,136	$6,401	$6,922	$6,392	$6,717	$6,521	$6,546	$6,643	$6,851
BRISTOL MYERS SQUIBB	$9,237	$9,531	$9,201	$9,011	$9,782	$9,492	$9,416	$9,377	$8,999
ELI LILLY & CO	$6,086	$7,026	$7,191	$9,350	$10,991	$13,337	$16,200	$18,279	$20,286
GSK	$5,910	$6,225	$6,259	$6,941	$7,697	$8,663	$9,238	$9,599	$9,730
JOHNSON & JOHNSON	$12,091	$14,589	$13,672	$14,243	$15,534	$14,702	$15,348	$16,159	$17,037
MERCK & CO	$9,231	$10,077	$11,842	$12,752	$14,265	$15,595	$16,600	$16,995	$17,725
NOVARTIS	$8,484	$9,041	$8,264	$8,740	$9,302	$10,295	$11,162	$11,736	$12,171
NOVO NORDISK	$2,368	$2,826	$3,405	$4,603	$6,971	$7,882	$7,503	$7,734	$8,180
PFIZER	$8,884	$10,344	$11,409	$12,158	$10,694	$10,212	$11,268	$11,400	$11,182
ROCHE	$12,960	$14,996	$14,725	$15,299	$14,814	$14,777	$15,645	$16,304	$17,082
SANOFI	$6,314	$6,732	$7,063	$7,340	$7,999	$8,865	$9,256	$9,674	$9,975
TAKEDA PHARMACEUTICALS	$4,433	$4,249	$4,671	$4,744	$5,053	$4,795	$4,406	$4,789	$5,041
OTSUKA HOLDINGS	$2,032	$2,116	$2,106	$1,990	$2,076	$2,360	$2,385	$2,509	$2,621
ASTELLAS PHARMA	$2,063	$2,118	$2,191	$2,010	$2,037	$2,151	$1,979	$2,138	$2,207
CSL	$922	$1,001	$1,156	$1,396	$1,430	$1,359	$1,223	$1,303	$1,353
UCB	$1,791	$1,927	$1,759	$1,774	$1,927	$2,060	$2,352	$2,561	$2,747
MERCK KGaA	$2,583	$2,860	$2,576	$2,570	$2,465	$2,730	$2,832	$2,930	$3,055
AMGEN	$4,085	$4,296	$4,341	$4,784	$5,878	$7,183	$7,401	$7,543	$7,697
BIOGEN	$3,991	$2,501	$2,231	$2,330	$1,930	$1,731	$1,945	$2,003	$2,009
GILEAD SCIENCES	$4,857	$5,226	$4,968	$5,721	$5,732	$5,687	$5,961	$6,198	$6,363
MODERNA	$1,370	$1,991	$3,295	$4,787	$4,543	$3,132	$2,949	$2,663	$2,332
Total Large Biopharma	**$129,529**	**$140,579**	**$145,182**	**$156,100**	**$167,663**	**$176,335**	**$186,195**	**$194,181**	**$201,640**
% Change Y/Y	15.0%	8.5%	3.3%	7.5%	7.4%	5.2%	5.6%	4.3%	3.8%

Note: Data as of 6/29/2026
Source: Deutsche Bank, Bloomberg Finance LP, Company Reports

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

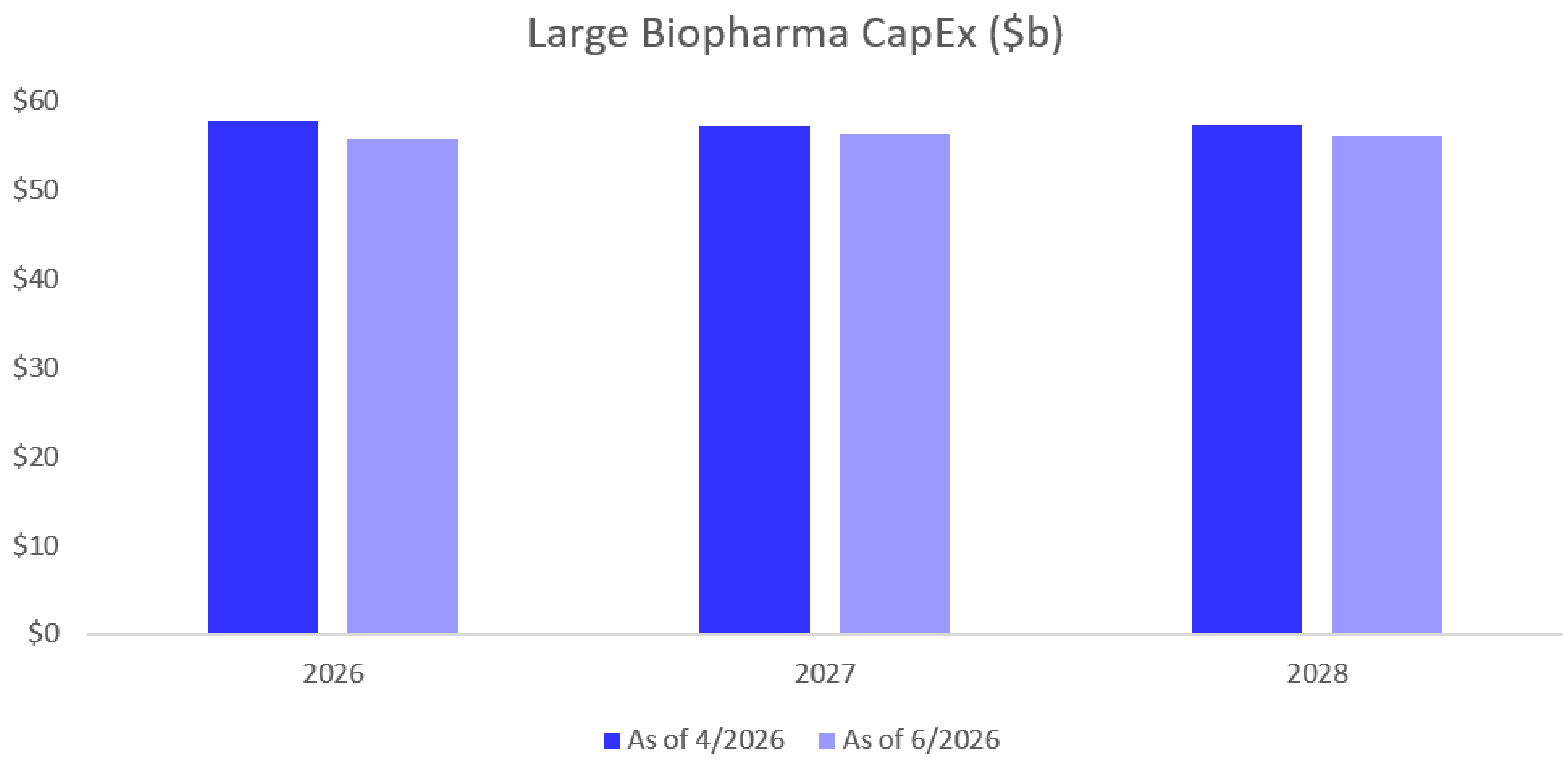
Large Biopharma R&D
Mid-single digit R&D growth through 2028



Note: Large Biopharma R&D Index includes 23 companies; 2026-2028 based on Bloomberg consensus forecasts as of 6/29/2026
Source: Deutsche Bank, Bloomberg Finance LP, Company Reports

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

Large Biopharma Capex Estimate Trends
Forward capex expectations modestly lower relative to 4/2026



Note: Group includes Bloomberg estimates for 23 large biopharma companies; estimates as of 6/29/2026
Source: Deutsche Bank, Bloomberg Finance LP, Company Reports

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

Large Biopharma Capex Trends

Total 2026E capex (3%) vs. 4/2026; Takeda, Astellas, and Bayer the culprits

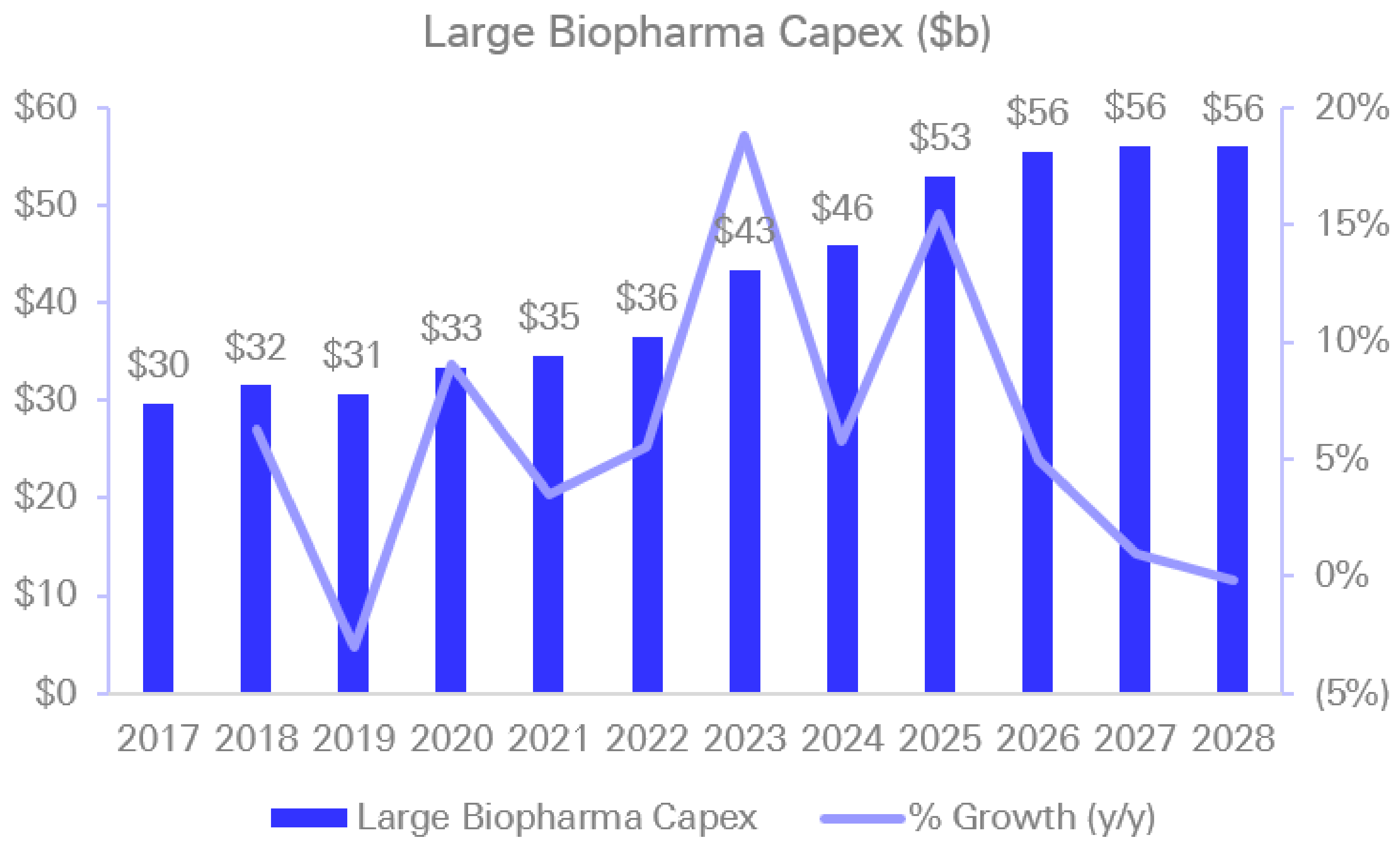
Large Biopharma Capital Expenditures	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
ABBVIE	$798	$787	$695	$777	$974	$1,214	$1,188	$1,539	$1,579
ASTRAZENECA	$961	$1,091	$1,091	$1,361	$2,053	$2,634	$3,292	$3,186	$3,225
BAYER	$2,761	$3,088	$3,106	$2,975	$3,005	$2,811	$3,154	$3,474	$3,570
BRISTOL MYERS SQUIBB	$753	$973	$1,118	$1,209	$1,193	$1,185	$1,237	$1,435	$1,373
ELI LILLY & CO	$1,388	$1,310	$1,854	$3,448	$4,141	$6,264	$6,923	$6,395	$6,140
GSK	$1,574	$1,307	$1,413	$1,635	$1,788	$1,778	$1,936	$2,041	$2,120
JOHNSON & JOHNSON	$3,347	$3,652	$4,009	$4,543	$3,851	$4,255	$4,840	$5,054	$5,442
MERCK & CO	$4,429	$4,448	$4,388	$3,863	$3,372	$3,167	$3,510	$3,878	$3,723
NOVARTIS	$1,275	$1,378	$1,198	$1,060	$1,213	$1,580	$1,746	$1,931	$2,172
NOVO NORDISK	$892	$1,008	$1,720	$3,746	$6,840	$9,109	$8,463	$6,032	$4,557
PFIZER	$2,252	$2,711	$3,236	$3,907	$2,813	$2,548	$2,388	$2,571	$2,494
ROCHE	$3,762	$4,040	$3,614	$4,167	$4,008	$4,524	$5,197	$5,349	$5,819
SANOFI	$2,378	$2,416	$2,318	$3,271	$3,456	$3,999	$3,139	$3,164	$3,226
TAKEDA PHARMACEUTICALS	$1,169	$1,049	$1,098	$1,040	$1,215	$1,318	$1,169	$2,380	$2,421
OTSUKA HOLDINGS	$457	$478	$466	$647	$631	$588	$585	$491	$484
ASTELLAS PHARMA	$380	$296	$274	$270	$264	$243	$383	$704	$742
CSL	$1,207	$1,196	$1,079	$1,228	$862	$767	$716	$860	$984
UCB	$292	$334	$265	$257	$253	$318	$470	$534	$607
MERCK KGaA	$1,613	$1,261	$1,613	$1,954	$1,841	$1,792	$1,773	$1,823	$1,890
AMGEN	$608	$880	$936	$1,112	$1,221	$2,056	$2,612	$2,279	$2,404
BIOGEN	$425	$258	$240	$277	$244	$197	$205	$206	$220
GILEAD SCIENCES	$650	$579	$728	$585	$585	$560	$624	$768	$795
MODERNA	$67	$284	$400	$707	$1,073	$285	$241	$213	$220
Total Large Biopharma	**$33,439**	**$34,824**	**$36,860**	**$44,038**	**$46,899**	**$53,190**	**$55,790**	**$56,308**	**$56,205**
% Change Y/Y	0.0%	4.1%	5.8%	19.5%	6.5%	13.4%	4.9%	0.9%	(0.2%)

Note: Bloomberg estimates are adjusted for IPR&D estimates where applicable; estimates as of 6/29/26
Source: Deutsche Bank, Bloomberg Finance LP, Company Reports

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

Large Biopharma Capex Trends

Capex up +mid-single digit in 2026



Source: Deutsche Bank, Bloomberg Finance LP, Company Report

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

Biopharma Investment Activity
Large biopharma investment activity spans both domestic and Chinese announcements

Company	Investment	Length of Plan	New Facilities	New Jobs	Modalities / Function
Eli Lilly	$4.5 billion	Unclear	N/A	N/A	Lilly announced an additional $4.5b investment across 2/3 Lebanon, IN sites
Amgen	$300 million	N/A	N/A	N/A	$300mm in U.S. manufacturing network to advance production capacity
AbbVie	N/A	N/A	1	N/A	$1.4b for NC manufacturing campus, part of AbbVie's $100b U.S. commitment
Novartis	Unclear	Unclear	1	N/A	56,200sf API manufacturing facility for solid dosage tablets, capsules, and RNA therapeutics
AstraZeneca	Unclear	Unclear	N/A	N/A	$3.0b in China manufacturing operations for oral solid drugs, specifically around GLP-1s
Novartis	$480 million	N/A	2	N/A	$480mm tied to 2 existing China facilities in Shanghai and Beijing
Eli-Lilly	$3.0 billion	10 years	N/A	N/A	$3.0b in China manufacturing operations for oral solid drugs, specifically around GLP-1s
AbbVie	$100 billion	10 years	~4	>6000	Expands U.S manufacturing footprint (API +biologics) and capacity investment
Amgen	~$2.5 billion	2 years	3	~1,120	U.S manufacturing capex (expansions) tied to broader MFN actions
AstraZeneca	$50 billion	5 years	~6	"tens of 000s"	U.S manufacturing expansion (incl.drug substance) + U.S hub/portfolio support (incl.weight-management)
Biogen	$2 billion	~3 years	<2	n/a	Multi-modality manufacturing footprint expansion (broad capacity build)
Bristol Myers	$40 billion	~5 years	n/a	n/a	U.S R&D and domestic manufacturing
Eli Lilly	$27 billion	5 years	4	>13,00	Metabolic, sterile injectables and other pipelines
Gilead	$32 billion	5 years	6	>3,000	Labs, R&D sites, Tech/Engineering upgrades and Manufacturing, + 3 site upgrades
GSK	$30 billion	5 years	2	n/a	R&D, Supply chain; new biologics capacity
J&J	$55 billion	4 years	4	>5,000	Innovative med. Biologics and MedTech Advanced Manufacturing
Merck	>$70 billion	5 years	3	~16,200	Vaccines, small-molecule Drug Manufacturing
Moderna	>$140 million	1H27	1	"00s"	End-to-end U.S mRNA manufacturing
Novo Nordisk	$10 billion	5 years	4	>12,000	Drug / Diagnostic manufacturing, CGT facility, Obesity programs
Novartis	$23 billion	5 years	10	c5,000	Biologics Drug Products, Drug Substance, Assembly, Oral Solid Dosage, Radioligand Therapy
Pfizer	$70 billion	2-5 years	n/a	n/a	U.S research, development and capital projects
Regeneron	$7 billion	10 years	2	<1,000	Preclinical manufacturing, Fill/ Finish
Roche	$50 billion	5 years	4	>12,000	Drug / Diagnostic manufacturing, CGT facility, Obesity programs
Sanofi	$20 billion	5 years	n/a	n/a	R&D, Manufacturing investment and partnerships with domestic manufacturers
Takeda	$30 billion	5 years	n/a	n/a	Maintaining / upgrading U.S manufacturing network and broader U.S operations

Source: Deutsche Bank, Company Reports

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

Biopharma U.S. Investment Activity

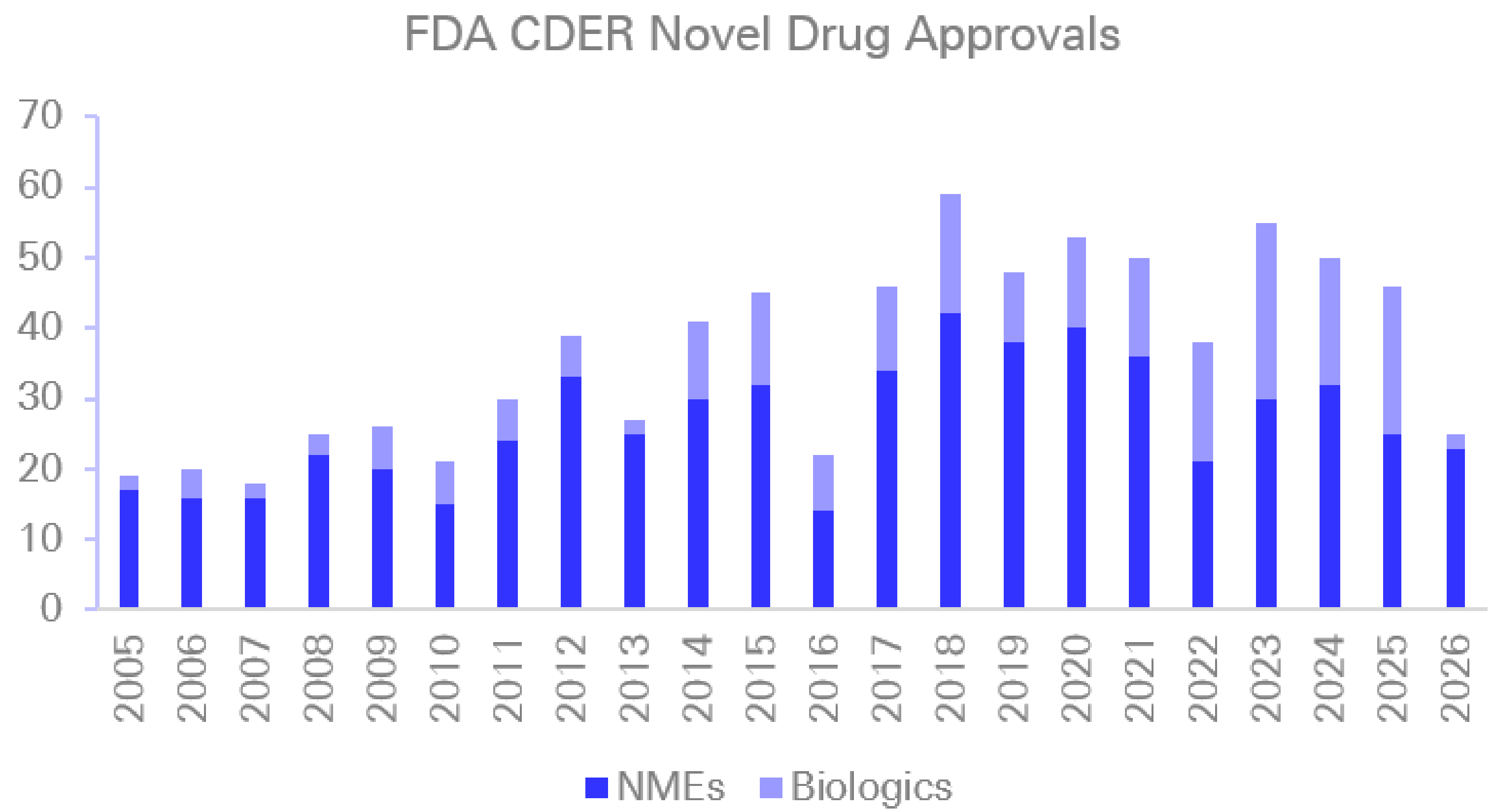
Large biopharma is highlighting U.S. investment plans as part of trade deals

Company	Location	Investment	Announced	New Sites	New Jobs	Modalities / Function
Eli Lilly	Lebanon, IN	$4.5b	6-May-26	Expansion	N/A	Lilly announced an additional $4.5b investment across 2/3 Lebanon, IN sites
J&J	Montgomery County, PA	>$1b	18-Feb-26	1 (new)	~4,500	Cell Therapy manufacturing (advanced medicines across oncology / neuro)
Eli Lilly	Lehigh Valley, PA	$3.5b	30-Jan-26	1 (new)	850 permanent + 2,000 construction	Weight-loss therapies (incl. retatrutide); GIP, GLP-1 and glucagon triple hormone receptor agonist
Roche / Genentech	Holly Spring, NC	$2b	20-Jan-26	1 (new)	>2,2000	Biomanufacturing for metabolic / obesity pipeline
Abbvie	Tempe, Arizona	$175mm	12-Jan-26	1 (new)	~200	Drug-delivery device manufacturing (incl. 3.5mL on-body injector); supports immunology +neuroscience
Eli Lilly	Huntsville, Alabama	$6b	9-Dec-25	1 (new)	450 permanent + 3,000 construction	Small-molecule & peptide API manufacturing (incl. oral GLP-1 orforglipron)
Eli Lilly	Houston, Texas	$6.5b	23-Sep-25	1 (new)	615 permanent + 4,000 construction	Next-gen small-molecule API plant (pipeline across cardiometabolic, oncology, etc.; will produce orforglipron)
Eli Lilly	Goochland,VA	$5b	16-Sep-25	1 (new)	650 permanent + 1,800 construction	Integrated API & drug product facility for bioconjugates
Eli Lilly	Carolina, Puerto Rico	$1.2b	29-Oct-25	Expansion	100 permanent + 1,000 construction	Expansion of oral solid-dose manufacturing (small-molecule pills for cardiometabolic, neuroscience, etc., incl. GLP-1 obesity pill)
AbbVie	North Chicago, Illinois	$195mm	12-Aug-25	Expansion	~50 new (per IL EDGE agreement)	New state-of-the-art API facility to expand U.S. chemical synthesis (immunology, oncology, neuroscience drugs)
AbbVie	Worcester, Massachusetts	$70mm	30-Sep-25	Expansion	N/A (not disclosed)	Biologics manufacturing & R&D expansion (adds labs, warehouse and enables transfer of oncology biologics production to U.S.)
Amgen	New Albany, Ohio	$900mm	25-Apr-25	Expansion	+350 permanent (to ~750 total jobs)	Biologics drug product expansion (fill-finish and packaging capacity)
Amgen	Juncos, Puerto Rico	$650mm	26-Sep-25	Expansion	~750 combined (construction + high-skill mfg.)	Biomanufacturing campus expansion (new labs and pilot production to support early-stage biologics pipeline)
Amgen	Holly Springs, North Carolina	$1b	5-Dec-24	1 (new)	370 new permanent jobs	Second drug-substance plant for biologics (highly automated, to meet rising demand for Amgen's therapies)
J&J	Wilson County, North Carolina	$2b	1-Oct-24	1 (new)	~420–500 permanent + ~5,000 construction	State-of-the-art biologics manufacturing facility (oncology, immunology, neuroscience products; future-proof U.S. capacity)
Merck & Co.	Wilmington, Delaware	$1b	29-Apr-25	1 (new)	>500 permanent + ~4,000 construction	Launch & commercial production center for next-gen biologics (ADC therapies) and future U.S. home of Keytruda manufacturing
Merck & Co.	Elkton, Virginia	$3b	20-Oct-25	Expansion	>500 permanent + ~8,000 construction	Pharmaceutical manufacturing Center of Excellence (small-molecule API and drug product capacity to strengthen U.S. supply)
Gilead Sciences	Foster City, California (HQ)	Part of $32b U.S. program	7-May-25	3 new facilities	~800 direct jobs (+ ~2,200 indirect by 2028)	footprint – new Technical Dev. Center, new research building, and biologics production facility (boosting
Genentech (Roche)	Holly Springs, NC	~$2b	15-May-25	1 (new)	500 permanent + 1,500 construction	Large-scale biologics facility to support obesity/metabolic dis
Roche Diagnostics	Indianapolis, Indiana	up to $550mm (by 2030)	12-May-25	Expansion	Hundreds MFG + thousands construction	Expansion of Indy campus into a major hub for continuous glucose monitor (CGM) production (domestic diabetes device manufacturing)
Novartis	U.S.-wide (e.g. NC, FL, TX, CA)	$23b (5-yr U.S. plan)	10-Apr-25	s (7 new + 3 expar	~1,000 new Novartis jobs (+ ~4,000 indirect)	End-to-end manufacturing expansion (small-molecule APIs, biologics, cell/gene therapies, radioligand production) to make 100% of key medicines in USA
Biogen	Research Triangle Park, NC	$2b	21-Jul-25	1 (new)	N/A	New biologics production facility (expands NC footprint for MS/neuroscience pipeline manufacturing; bolsters U.S. drug supply amid tariff concerns)
Pfizer	Kalamazoo, MI & Sanford, NC	$0.75–1b+ (multi-phase)	21-Jul-25	Expansion	N/A	Multi-modality capacity boost – adding sterile injectable production lines (Kalamazoo) and gene therapy mfg. capacity (Sanford)
Regeneron	Tarrytown, NY (+ planned VA)	~$7b(through 2030)	1-Apr-25	expanding NY/VA	TBD	Broad R&D and manufacturing expansion (scaling antibody production in NY; potential new VA site; also partnering with CDMOs to secure U.S. biologics capacity)
AstraZeneca	Holly Springs, NC (via FDB)	$2b (10-year commitment)	21-Aug-25	Leased capacity	~120 jobs (at FDB site)	site – drug substance & fill-finish for AZ's advanced
Novo Nordisk	Clayton, North Carolina	$4.1b	26-Jun-24	1 (new)	1,000 new permanent (+2,000 construction)	API/formulation facility for obesity and diabetes drugs (e.g. GLP-1 therapies like Wegovy/Ozempic)
Sanofi	U.S. (multi-state)	>$20b (by 2030)	14-May-25	Multi-site	TBD	Domestic manufacturing + R&D expansion; includes biologics, mRNA, clinical trials scale-up

Source: Deutsche Bank, Company Reports

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

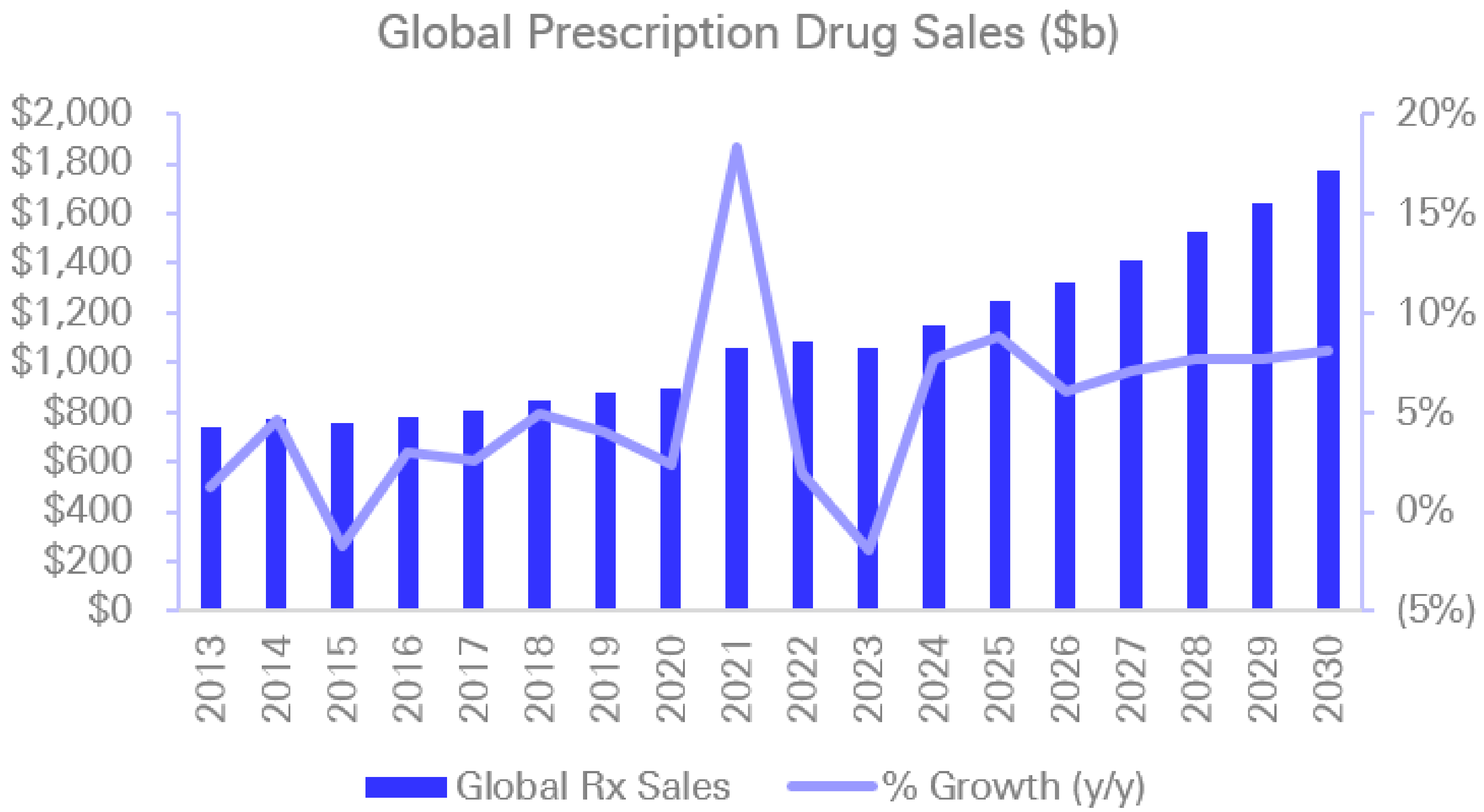
FDA approvals
Tracking to 50 approvals, up from 44 pre-1Q26



Note: Center for Drug Evaluation & Research (CDER) approves sm. molecule and biologics; Office of Tissues & Advanced Therapeutics (OTAT) approves C&GT products
Source: Deutsche Bank, FDA as of 6/30/2026

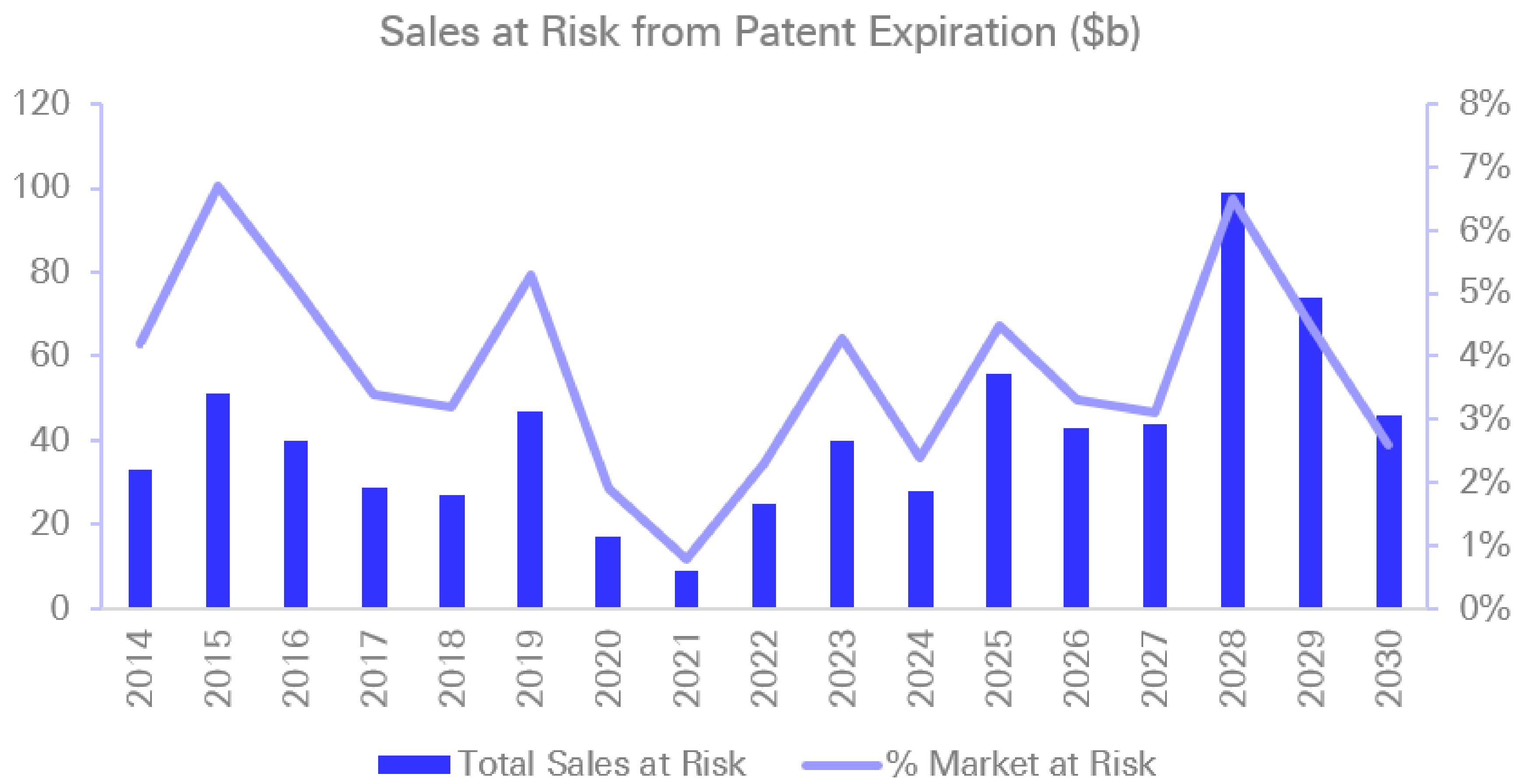
Justin Bowers; justin.bowers@db.com; +1.917.517.9836

Prescription Drug Sales
Mid-single digit growth in 2026



Source: Deutsche Bank, EvaluatePharma

Biopharma Patent Expiries
Industry facing >$300b of sales losing exclusivity from 2026-2030

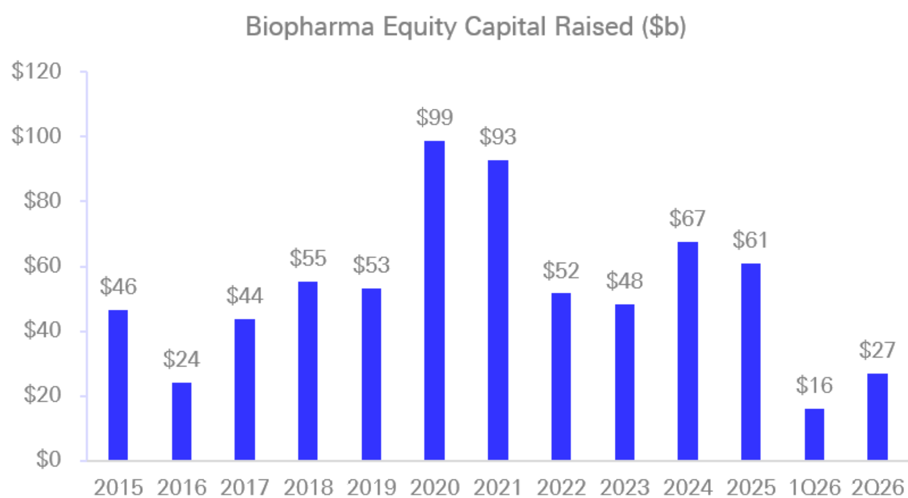


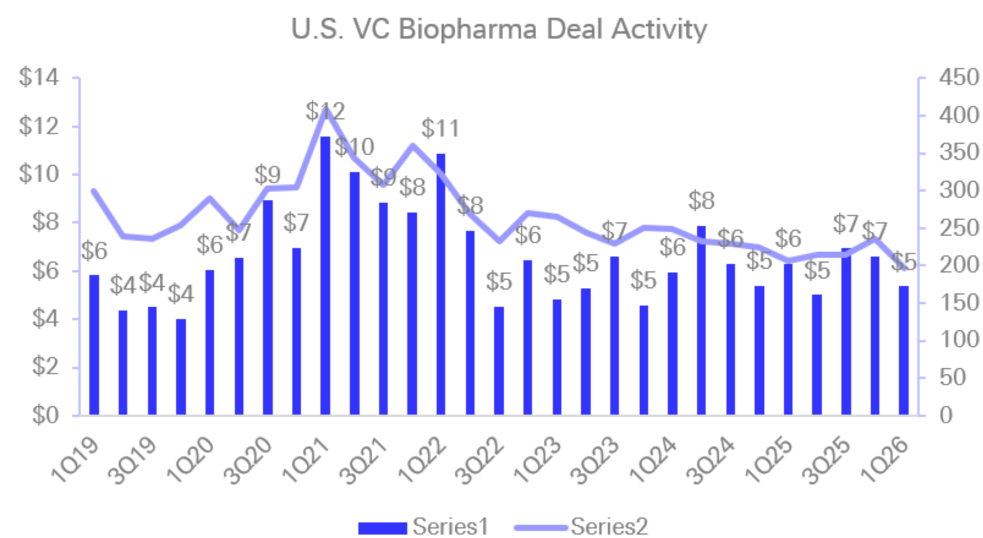
Source: Deutsche Bank, Bloomberg Finance LP, EvaluatePharma

Biotech equity capital improvement continues; on-pace to surpass 2025

— 2Q26 funding continues to show improvement with total biopharma equity at $27b, the third greatest quarterly amount on record and since 2020-2021; biopharma equity capital on pace for ~$80b raised based on the 1H26 run-rate (3rd greatest year)

— Initial data also supportive of early-stage / biotech VC equity improving off 1Q26 levels QoQ



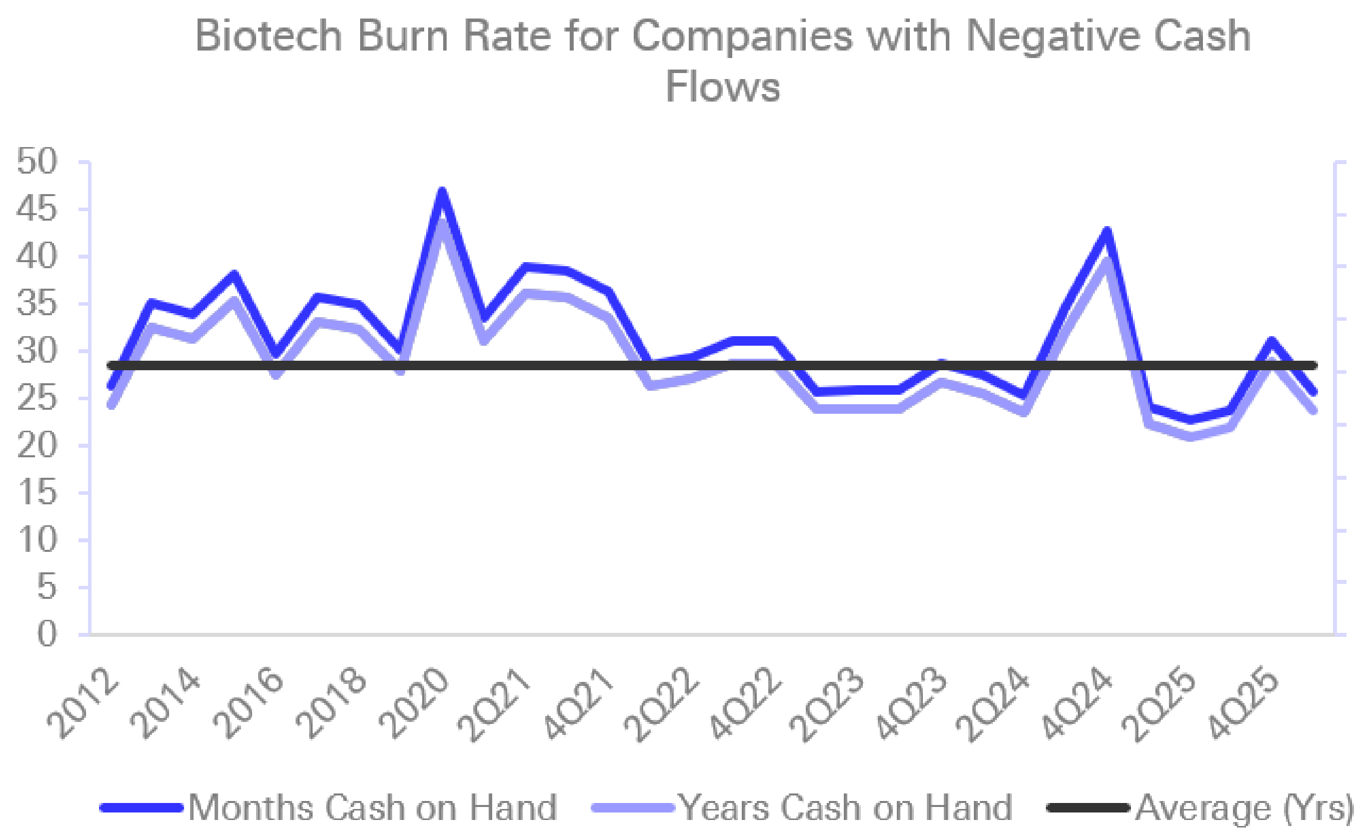


Source: Deutsche Bank, BioCentury, Bloomberg Finance LP, Crunchbase, PitchBook, NVCA

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

Biotech Burn Rate Tracker

Weakness relative to 4Q25 but still YoY improvement

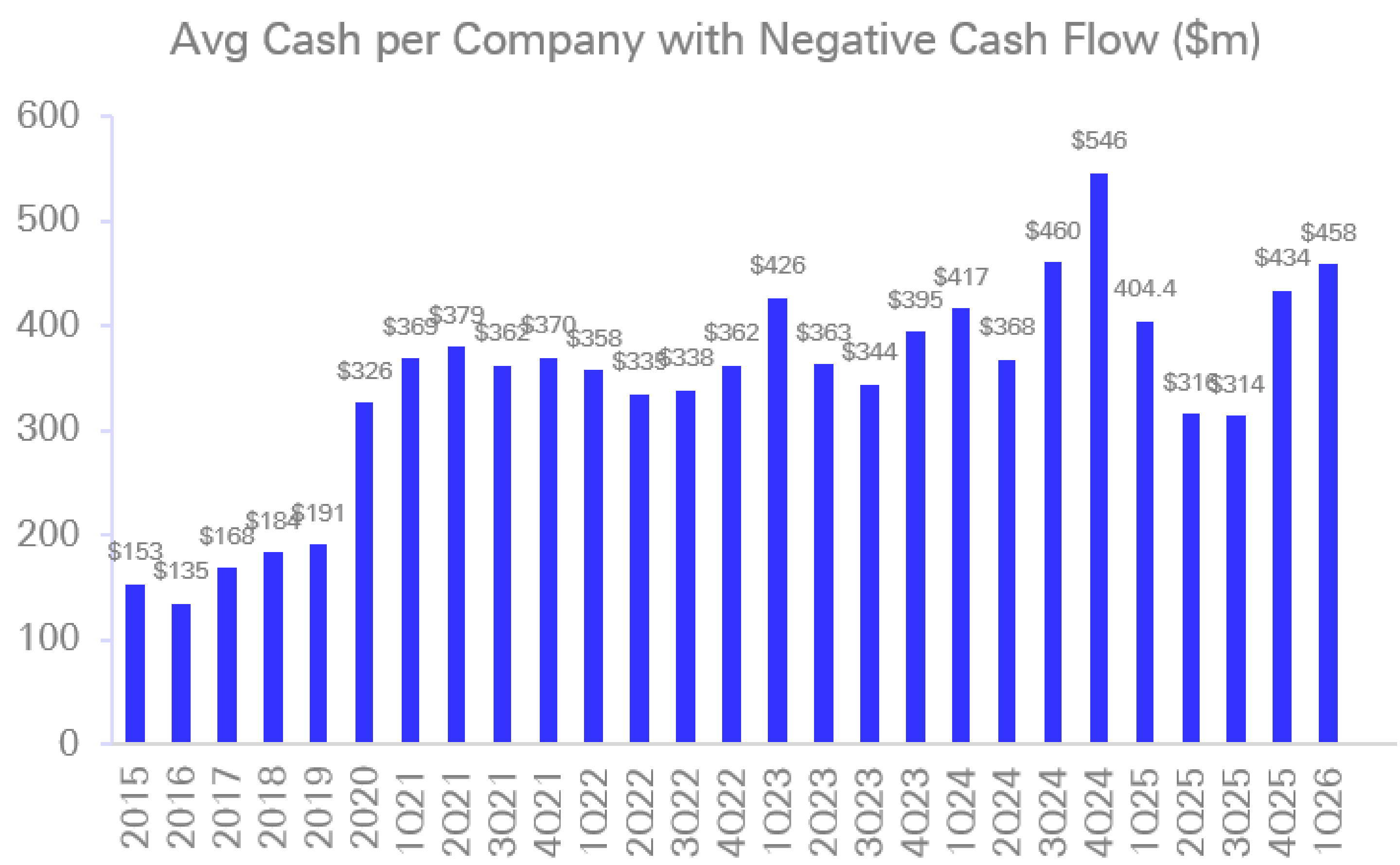


Note: The Biotech Burn Rate Tracker includes 372 public companies as of 4Q25

Source: Deutsche Bank, Bloomberg Finance LP, Company Reports

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

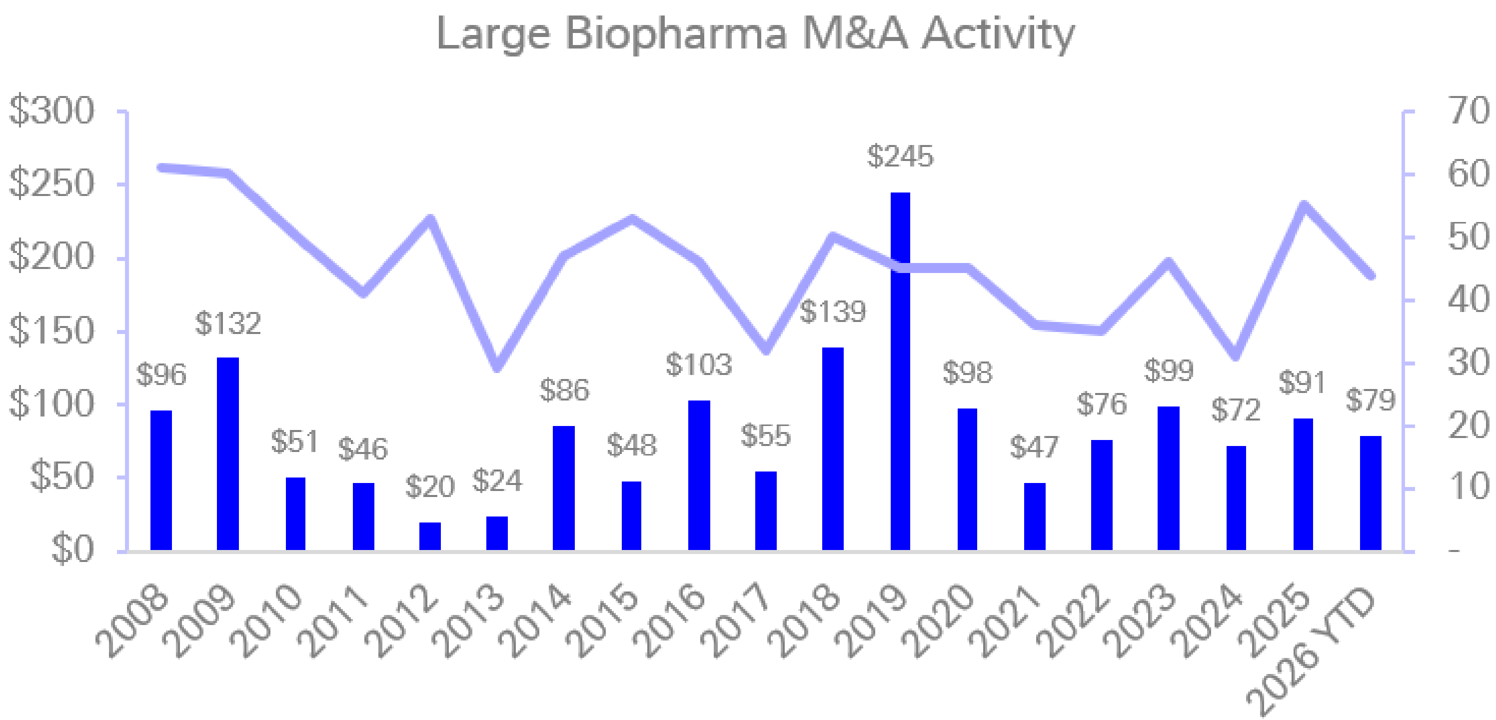
Biotech Burn Rate Tracker
QoQ recovery off 3Q25 lows continues



Note: The Biotech Burn Rate Tracker includes 372 public companies as of 1Q26
Source: Deutsche Bank, Bloomberg Finance LP, EvaluatePharma

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

Large Biopharma Strategies to Execute Sales Growth
YTD value / number of 2026 deals remains on-pace to eclipse 2025



Note: Group includes 23 large biopharma companies; 2025 / 2026 YTD include pending, announced, and completed deals
Source: Deutsche Bank, Bloomberg Finance LP

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

M&A Activity and Multiples Since 2023
Transaction volume and size picking up after 2024 in Tools / CRO / Dx

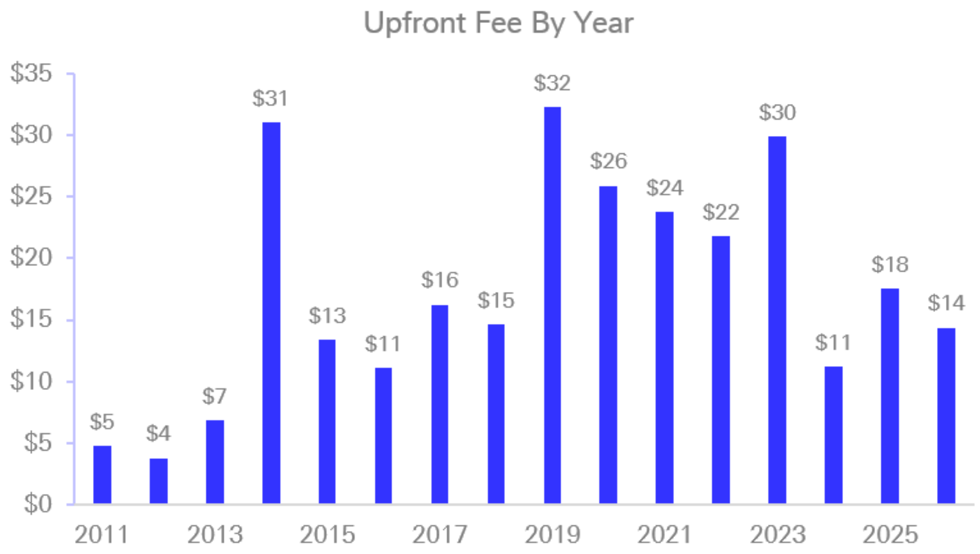
Tools / CRO M&A Acquirer	Target	Date Announced	Date of Completion	Value ($ mm)	Valuation Metric	Implied TTM Valuation
Merck KGaA	Bio-Techne	6/25/2026	TBD	$11,300	EV / EBITDA	26.5x
Agilent	Biocare Medical	3/9/2026	6/24/2026	950	EV / Sales	10.6x
Danaher	Masimo	2/17/2026	6/11/2026	9,669	EV / Sales	6.3x
Worldwide Clinical Trials	Catalyst	1/20/2026	N/A	~500	EV / EBITDA	20.0x
Abbott	Exact Sciences	11/20/2025	3/23/2026	21,400	EV / Sales	7.8x
Thermo Fisher	Clario	10/29/2025	3/24/2026	8,875	EV / Sales	7.1x
Blackstone / TPG	Hologic	10/21/2025	12/18/2024	17,500	EV / Sales	4.1x
IQVIA	Cedar Gate	10/8/2025	4Q25	750	EV / EBITDA	20.8x
Waters	BD Biosciences / Dx	7/14/2025	2/10/2026	17,500	EV / Sales	5.4x
Bain Capital	PCI Pharma	7/14/2025	N/A	10,000	N/A	N/A
Illumina	SomaLogic	6/23/2025	1/30/2026	425	N/A	N/A
TPG / GIC	Novotech	3/31/2025	N/A	N/A	EV / EBITDA	High-teens
Thermo Fisher	Solventum Filtration Business	2/25/2025	9/2/2025	4,100	EV / Sales	4.1x
Agilent	Biovectra	7/22/2024	9/23/2024	925	EV / Sales	9.3x
Bruker	Nanostring Assets	4/18/2024	5/6/2024	393	EV / Sales	2.3x
Bruker	ELITech Group	2/28/2024	5/2/2024	943	EV / Sales	6.3x
Novo Holdings	Catalent	2/5/2024	12/18/2024	16,500	EV / EBITDA	23.3x
Thermo Fisher	Olink Holdings	10/17/2023	7/11/2024	3,084	EV / Sales	22.0x
Repligen	Metenova	9/26/2023	10/2/2023	170	EV / Sales	8.1x
Danaher	Abcam	8/28/2023	12/7/2023	5,673	EV / Sales	13.0x
Bruker	Berkeley Lights	8/17/2023	10/3/2023	147	EV / Sales	2.0x
Kohlberg & Company	Worldwide Clinical Trials	8/7/2023	12/12/2023	2,000	EV / EBITDA	18.0x
Thermo Fisher	CorEvitas	7/6/2023	8/14/2023	913	EV / Sales	8.3x
Elliott & Others	Syneos Health	5/10/2023	9/28/2023	7,100	EV / EBITDA	10.2x

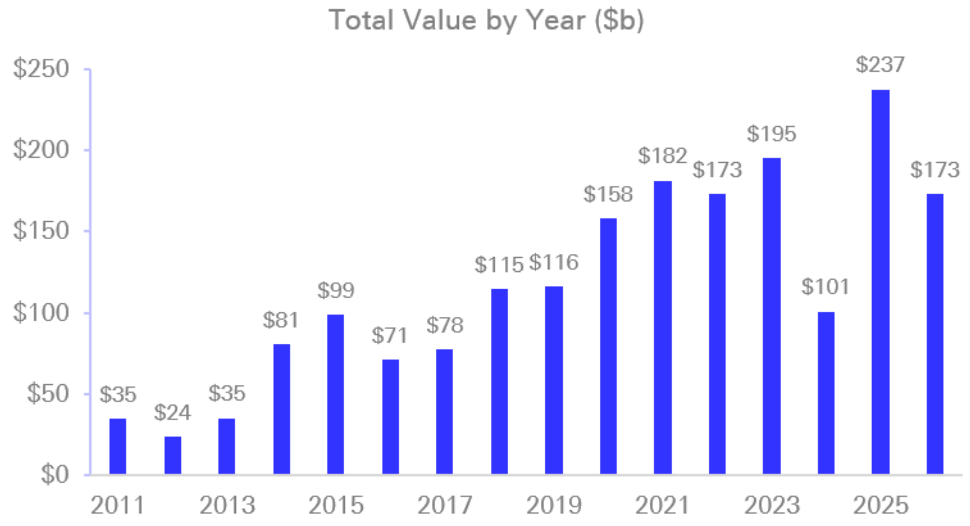
Source: Deutsche Bank, Bloomberg Finance LP, Company Report

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

2Q26 levels significantly improved YoY; 2026 already > 2024, on-track to easily clear 2025

- Through 1H26, total licensing deal value is on-track to easily surpass 2025 and has already surpassed 2024 with 6 months to go
- 2Q26 licensing deal value of $79b represents a +70% increase over 2Q25 levels of $47b; upfront fees of $5b +20% versus 2Q25 levels of $4b

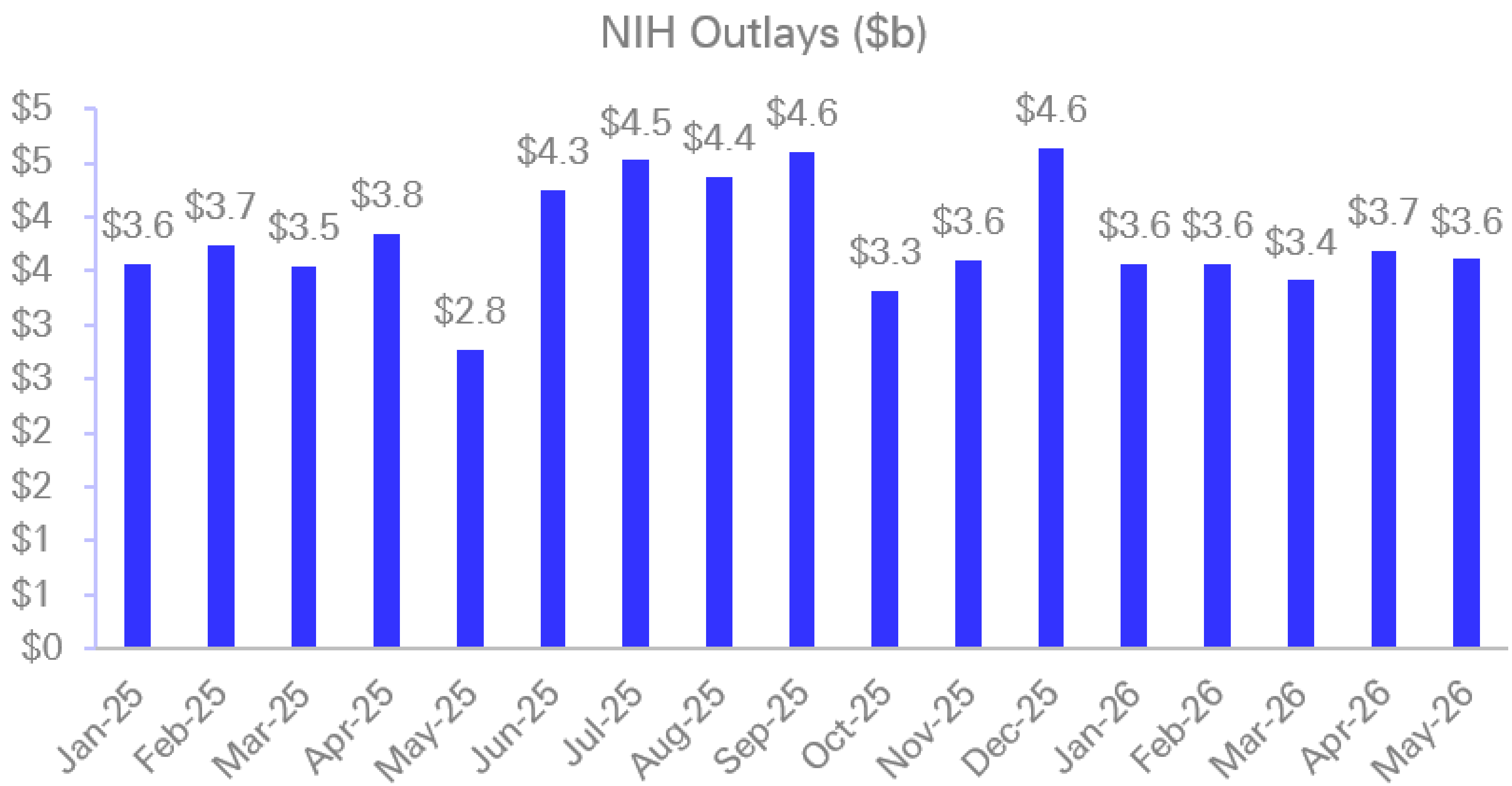




Note: As of 6/30/2026
Source: Deutsche Bank, Bloomberg Finance LP

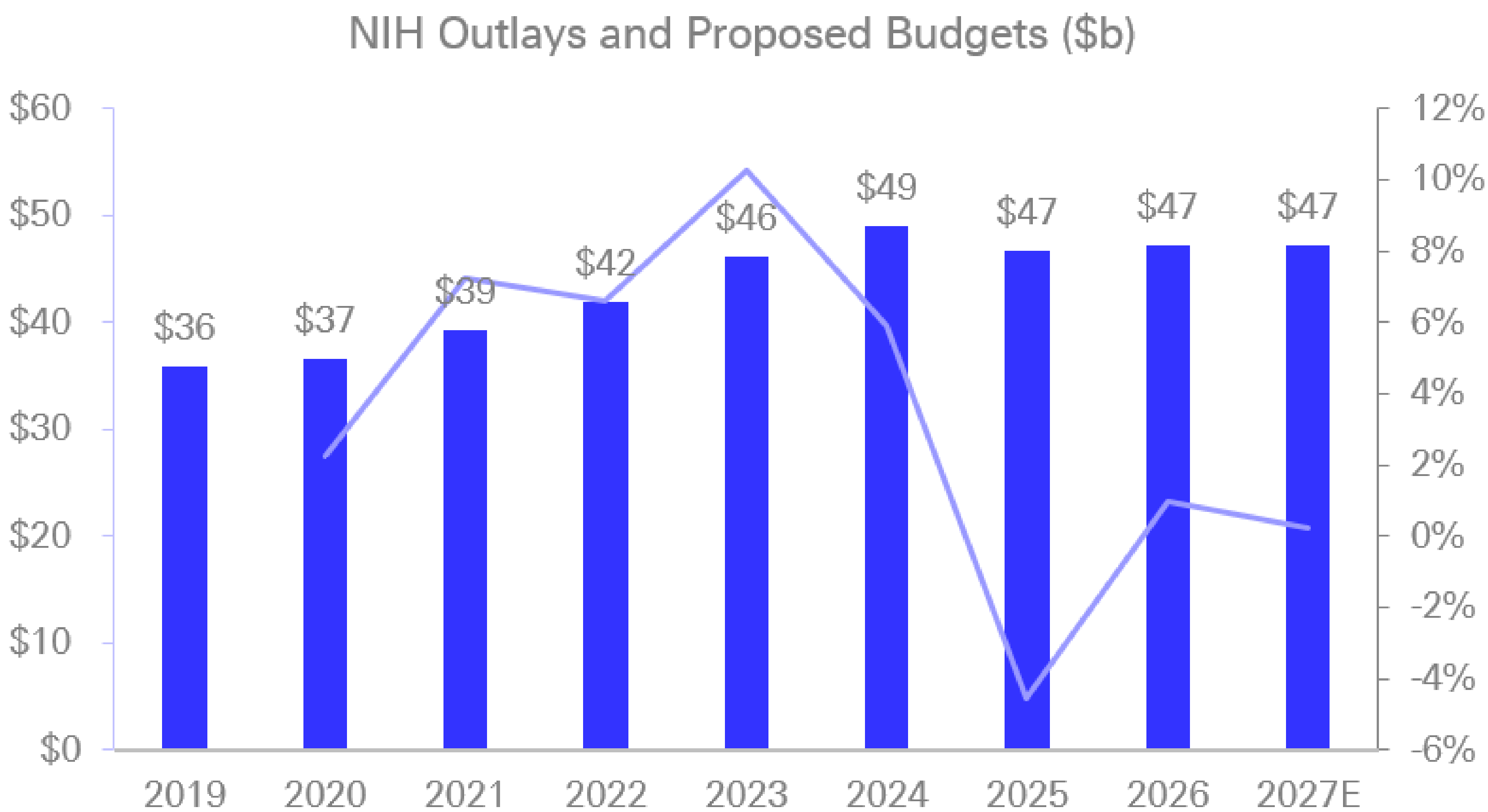
Justin Bowers; justin.bowers@db.com; +1.917.517.9836

NIH Outlays
Outlay +30% YoY in May 2026; +10% QTD vs. April / May 2025



Source: Deutsche Bank, NIH

NIH Funding
2026 budget flat YoY; proposed +$100mm increase in F2027



Source: Deutsche Bank, NIH

Worldwide A&G Funding Developments

U.S. A&G currently stable, but policy remains volatile; RoW investing across sciences

— White House proposed a ($5b) or (11%) YoY reduction vs. 2026 NIH budget; House rejected this and is proposing a +$100mm increase to $47.3b for the 2027 NIH budget

— The United States' proposed changes to the federal government's Uniform Guidance include requirements for senior political appointees to review discretionary grant award for alignment with administration priorities and executive orders

— China's Natural Science Fund budget is set at 42b yuan (about $6b) for 2026, an increase of 2b yuan from the 2025 spending level / YoY growth of +6%; funding will support 18 categories of scientific programs, including general research projects, youth science fund programs, regional science funds and innovation research groups.

— Japan has approved a five-year strategy starting fiscal 2026 and emphasizing science / technology policy; the plan calls for a record 60t yen in government R&D investment / raising Japan's global ranking in top-tier research papers from 13th to 3rd within a decade.

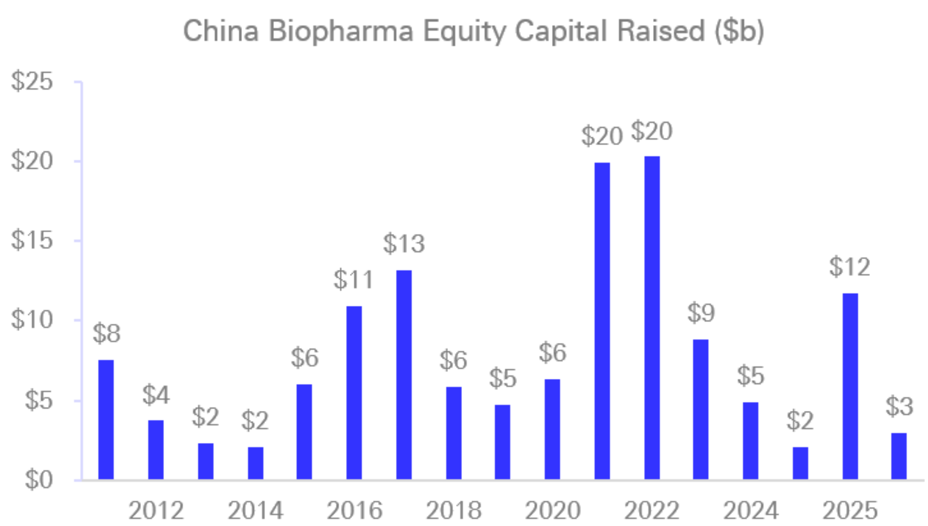
— From 2026 - 2030, the South Korean government plans to invest more than 200t won in research and development (R&D); 60t won will be concentrated in ten national strategic technology areas including artificial intelligence (AI), semiconductors, advanced bio, and quantum technologies.

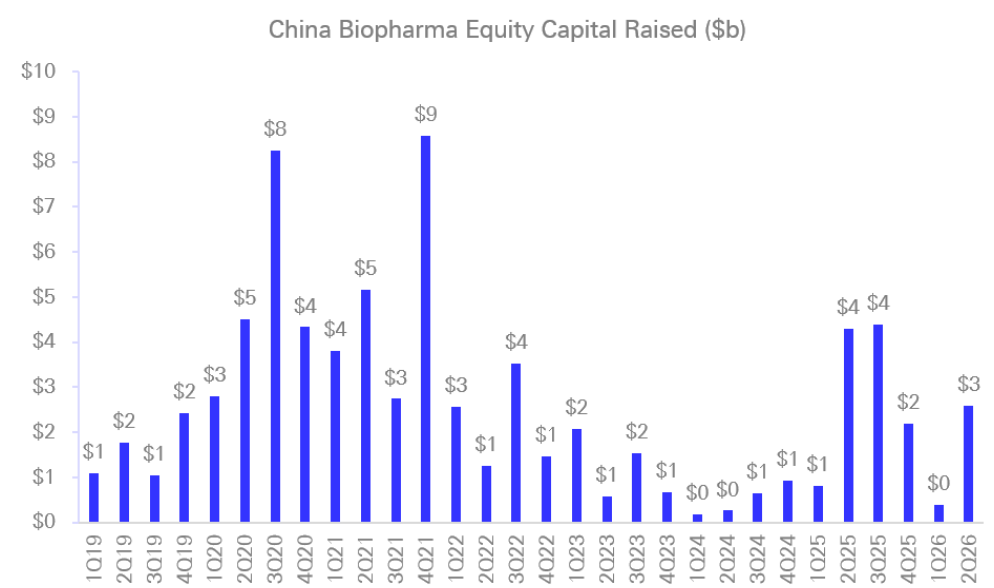
— The EU council has agreed on a partial negotiating position for regulation / specific programs of Horizon EU clearing the two legislative proposals of the package while the financial / horizontal issues will be discussed with the next MFF covering the period 2028 - 2034

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

China Capital Funding
2Q26 down YoY; but QoQ improvement particularly in Hong Kong

- 2Q26 off (40%) YoY, but represented a QoQ improvement particularly in Hong Kong where activity was absent in 1Q26
- Industry experts have indicated increased preclinical / early-stage work outsourced to China / India (+5 – 10% more work vs. 2 -3 years ago); work at this stage done in China / India is (35 – 20%) cheaper / +40% faster than in the U.S.





Source: Deutsche Bank, BioCentury, Bloomberg Finance LP

State of the Union on Life Science Tools / CROs

A quick overview of relevant management changes and special situations in the space

Company	CEO	CFO	Special Situation	Notes
10X Genomics				Commercial realignment
Avantor	X	X	X	New CEO this year, undergoing ongoing review of operations, new CFO
Bio-Techne			X	Merck KGaA acquiring for $73 / share
Charles River	X	X	X	New CEO
Concentra				
Danaher		X		CFO transition occuring in 2026
Fortrea	X	X	X	New CEO in 2025 and new strategic framework initiated, new CFO
ICON	X	X		New CEO in 2025, new CFO in 4Q24, accounting probe resolved
IQVIA		X		CFO transition occuring in 2026
Maravai	X	X	X	CEO / CFO transition in 2025, announcement of strategic review and initiatives
Medpace				President transition
Repligen				
Select Medical		X	X	Take-private approved in 1Q26
Thermo Fisher		X		CFO transition occuring in 2026
Waters			X	Becton Dickinson Life Sciences M&A closed in 1Q26
West Pharma	X	X		CEO transition in 2026, CFO transition in 2025

Source: Deutsche Bank, Company Reports

Justin Bowers: justin.bowers@db.com: +1.917.517.9836

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

2H26 margin cadence; AI-based revenue; bookings / demand metrics; and EBP demand

— 3Q / 4Q26 margins are expected to decelerate relative to 1H26. What line of sight does Iqvia have into the 2H26 margins and what levers is it pulling to ensure it can deliver on its margin targets?

— Iqvia is now employing 192 AI-based agents (up from 150 in 4Q25), what is the current revenue from these 192 agents, and ultimately what sort of AI-based opportunity is Iqvia targeting?

— The company has not lost a single trial to any AI-based alternatives, is these any market share erosion from AI contemplated in the company's internal constructs in 2027+?

— B2B of 1.04x in RDS up from 1.02X in 1Q25 with some weakness stemming from lower passthroughs and clinical trial mix, what was the clinical trial composition of bookings in the quarter and what type of therapeutic target was the weaker book-to-bill attributed to?

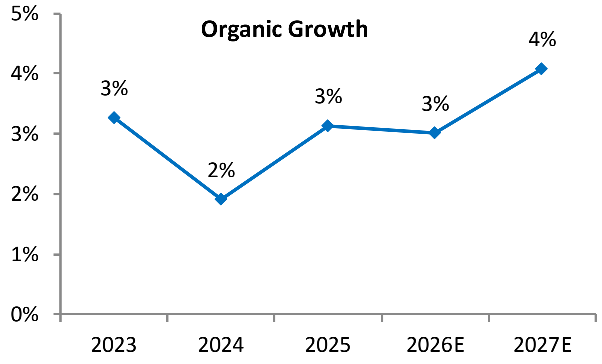
— RFPs +high-single-digit and qualified pipeline +mid-single-digit YoY, how did these metrics trend QoQ? What sort of annual book-to-bill expectations do these metrics support?

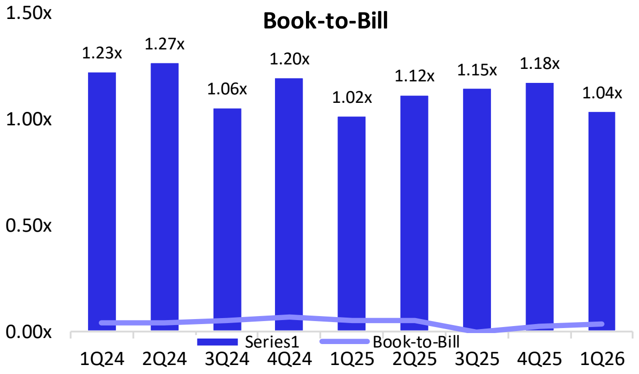
— Large pharma remains stabilized, albeit more constructive on assets closer to commercialization, while EBP continues to improve. Can investors expect improved funding / EBP to translate to bookings / improved top line growth by 4Q26?

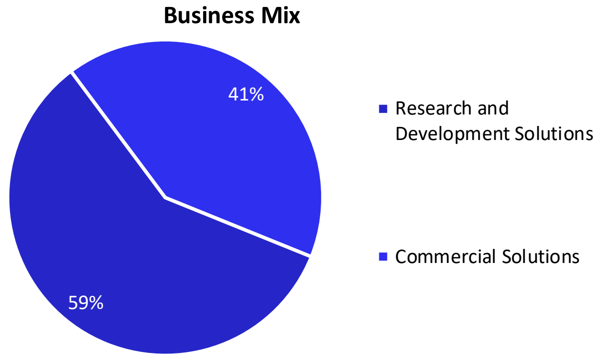
Justin Bowers; justin.bowers@db.com; +1.917.517.9836

IQVIA (IQV) / Buy / $240 Price Target

Company Overview







Financial Trends

IS Items ($ millions)	2023	2024	2025	2026E	2027E
Total Revenue	14,984	15,405	16,310	17,158	18,045
yoy chg.	*4.0%*	*2.8%*	*5.9%*	*5.2%*	*5.2%*
EBITDA	3,569	3,684	3,788	3,960	4,193
yoy chg.	*6.7%*	*3.2%*	*2.8%*	*4.5%*	*5.9%*
EBIT	3,017	3,087	3,169	3,326	3,537
yoy chg.	*8.9%*	*2.3%*	*2.7%*	*5.0%*	*6.3%*
EPS	$10.21	$11.13	$11.92	$12.62	$14.00
yoy chg.	*0.4%*	*9.1%*	*7.0%*	*5.9%*	*10.9%*

BS Items ($ millions)	2023	2024	2025	2026E	2027E
Cash	1,376	1,702	1,980	2,369	2,039
Total Debt	13,673	13,983	15,724	16,920	16,920
Net Debt	12,297	12,281	13,744	14,551	14,881
CF Items ($ millions)					
CFO	2,149	2,716	2,654	2,605	2,994
Free Cash Flow	1,500	2,114	2,051	2,012	2,396
Share Repurchases	(1,053)	(1,350)	(1,244)	(1,752)	(1,600)
Leverage Ratios					
Net Debt/EBITDA	3.4x	3.3x	3.6x	3.7x	3.5x

Valuation & Risks

Valuation

Price	$207.85
Market Cap ($M)	34,690
DB Rating	Buy
Valuation Methodology	13x 2027 EV/EBITDA
Price Target	$240.00

Risks

Risks to our thesis include: 1) large pharma pipeline prioritizations; 2) R&D spending cuts; 3) weak biotech funding environment; 4) decline in outsourcing penetration; 5) pricing pressure; 6) market share losses; 7) regulatory and geopolitical risk; and/or, 8) margin pressure.

Valuation Methodology

Our price target for IQV stock is based on 13x our 2027 EBITDA estimate, a discount with the stock's five-year trading average with improving end-market conditions offset by a contraction in industry multiples.

Source: Deutsche Bank, Company Filings, Bloomberg Finance LP
Note: Price & Mkt Cap as of 07/09/2026

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

MEDP: Top Questions
Cancelation trends, win-rate drivers; AI efficiencies; and forward growth rates

— What are the drivers of the elevated cancelation rates which are now at the highest level in over a year? Medpace cited futility as a driver; how are these cancellations dispersed across certain modalities, e.g., cardiometabolic, oncology, CGT etc. and by customer size?

— With concerns around the go forward growth rate driven by the elevated cancelation rates. Beyond bookings metrics required to underwrite the 15% a year growth assumption, what sort of cancelation rates would underwrite a 15% YoY increase, a +10% YoY increase, etc.?

— How have the recent elevated cancelation rates been incorporated into forward assumptions given guidance was maintained? Are cancelations normally reflected 4 quarters out? 6 quarters out?

— What is the relevant timeframe that Medpace can leverage an increase in bookings to counteract the increased cancelations? If 2 straight quarters of above-average bookings occur, is it enough to reverse weak forward assumptions driven by increased cancelations?

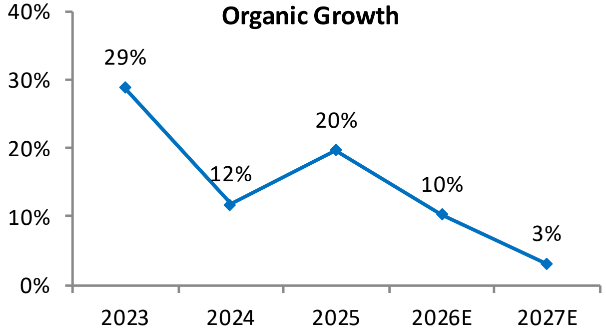
— AI still not expected to be a benefit for 2+ years, and in the near-term costs are expected to outweigh any efficiency gains. How is management weighing expected benefits from AI vs. efficiency gains to customers?

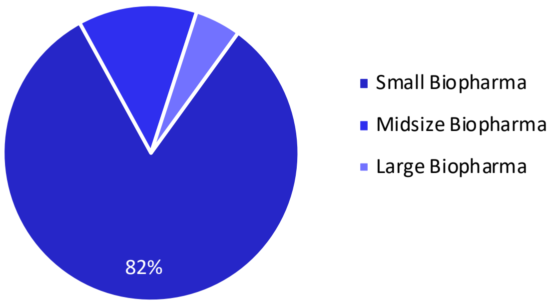
— Referenced initiatives being implemented to improve win-rates among customers, what are the details? Is it further discounting in an already price sensitive environment or more flexible study arrangements?

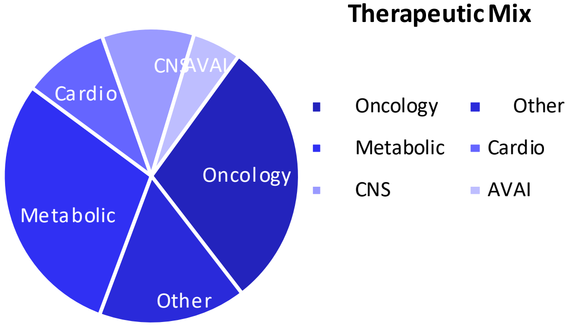
Justin Bowers; justin.bowers@db.com; +1.917.517.9836

Medpace (MEDP) / Hold / $480 Price Target

Company Overview







Financial Trends

IS Items ($ millions)	2023	2024	2025	2026E	2027E
Total Revenue	1,886	2,109	2,530	2,794	2,880
yoy chg.	*29.2%*	*11.8%*	*20.0%*	*10.4%*	*3.1%*
EBITDA	362	480	558	606	657
yoy chg.	*17.7%*	*32.5%*	*16.2%*	*8.6%*	*8.4%*
EBIT	337	447	535	577	629
yoy chg.	*20.9%*	*32.7%*	*19.7%*	*7.9%*	*9.0%*
EPS	$8.89	$12.63	$15.28	$17.05	$19.19
yoy chg.	*22.0%*	*42.2%*	*20.9%*	*11.6%*	*12.5%*

BS Items ($ millions)	2023	2024	2025	2026E	2027E
Cash	245	669	497	1,061	1,237
Total Debt	0	0	0	0	0
Net Debt	(245)	(669)	(497)	(1,061)	(1,237)
CF Items ($ millions)					
CFO	433	609	713	584	653
Free Cash Flow	397	572	682	553	627
Share Repurchases	(144)	(170)	(917)	0	(450)
Leverage Ratios					
Net Debt/EBITDA	(0.7x)	(1.4x)	(0.9x)	(1.8x)	(1.9x)

Valuation & Risks

Valuation

Price	$535.50
Market Cap ($M)	15,294
DB Rating	Hold
Valuation Methodology	28x 2026 EPS
Price Target	$480.00

Risks

Upside risks: robust biotech funding environment, market share gains, margin expansion, and capital deployment. Risks to the downside include margin compression from employee growth, scale challenges and market share losses from competitive disruption, slowing fundraising and R&D spend by the development stage biotech sector, and slowing R&D spend due to customer concentration.

Valuation Methodology

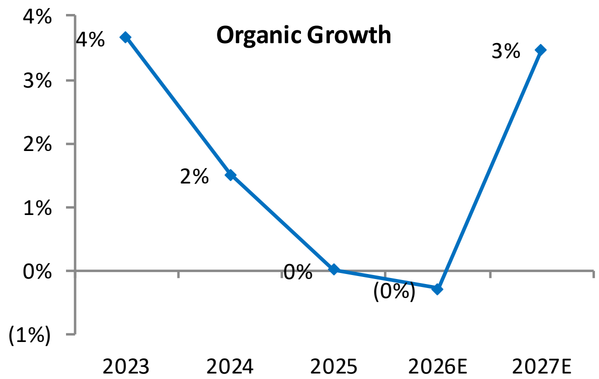
Our target multiple of 28x 2026E EPS is in-line with the 5YA forward multiple, reflective of recovering sector multiples / growth.

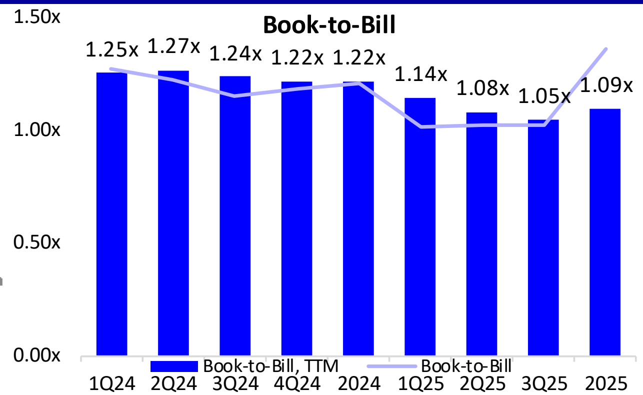
Source: Deutsche Bank, Company Filings, Bloomberg Finance LP
Note: Price & Mkt Cap as of 07/09/2026

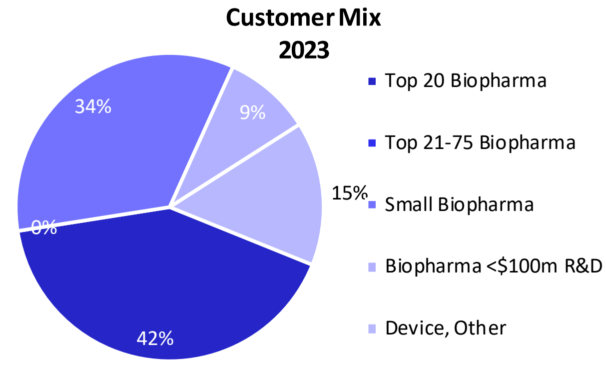
Justin Bowers; justin.bowers@db.com; +1.917.517.9836

ICON (ICLR) / Buy / $188 Price Target

Company Overview







Financial Trends

IS Items ($ millions)	2023	2024	2025	2026E	2027E
Total Revenue	8,055	8,189	8,251	7,996	8,220
yoy chg.	*4.0%*	*1.7%*	*0.8%*	*(3.1%)*	*2.8%*
EBITDA	1,643	1,670	1,531	1,319	1,435
yoy chg.	*11.0%*	*1.7%*	*(8.4%)*	*(13.8%)*	*8.7%*
EBIT	1,533	1,556	1,381	1,159	1,268
yoy chg.	*11.7%*	*1.5%*	*(11.2%)*	*(16.1%)*	*9.4%*
EPS	$12.08	$13.37	$12.52	$10.46	$11.66
yoy chg.	*2.9%*	*10.7%*	*(6.4%)*	*(16.4%)*	*11.5%*

BS Items ($ millions)	2023	2024	2025	2026E	2027E
Cash	378	539	647	1,252	2,062
Total Debt	3,776	3,426	3,402	3,374	3,345
Net Debt	3,397	2,887	2,755	2,122	1,283
CF Items ($ millions)					
CFO	1,161	1,287	1,036	843	1,039
Free Cash Flow	1,020	1,119	862	646	839
Share Repurchases	0	(500)	(750)	0	0
Leverage Ratios					
Net Debt/EBITDA	2.1x	1.7x	1.8x	1.6x	0.9x

Valuation & Risks

Valuation

Price	$168.77
Market Cap ($M)	12,928
DB Rating	Buy
Valuation Methodology	11x 2027 EV/EBITDA
Price Target	$188.00

Source: Deutsche Bank, Company Filings, Bloomberg Finance LP
Note: Price & Mkt Cap as of 07/09/2026

Risks

imm

Valuation Methodology

Our price target for ICLR stock reflects a multiple of 11x our 2027 EBITDA, a significant discount to the stock's historical average trading range to reflect uncertainty related to Icon's revenue recognition issue, AI disintermediation fears, and, contraction in market multiples.

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

2H26 margin outlook; segment growth expectations; bookings metrics; and NAMs TAM

— Guidance calls for >+500bps of operating margin expansion. With >1/2 driven by M&A and announced divestitures driving ~250 bps each. What's in the company's control over the remaining levers required to hit the required margin expansion and/or upside potential?

— Segment headwinds ease in 2Q26, what do easing headwinds mean for the range of organic growth investors can underwrite in each segment in 2Q26?

— 1.04x book-to-bill basically flat YoY and EBP book-to-bill and bookings the last 2 quarters at the highest level in >2 years. How should investors be modeling the 4Q26 exit rate book-to-bill with EBP bookings improving?

— Bookings trends support a >1.0x DSA net book-to-bill and positive DSA growth in 2H26. What is line of sight into this outlook and is the current backlog capable of supporting organic DSA growth in 2H26?

— Are shortened timelines for decision making at large pharma clients indicative of demand trends shifting back in CRO's favor?

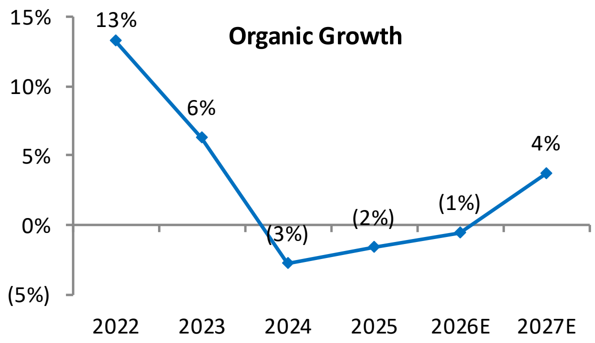
— Among EBP the new norm seems to be a higher burden of proof required to receive funding / move trials forward, does it also mean the EBP backlog is healthier given the trials moving forward have more / better data?

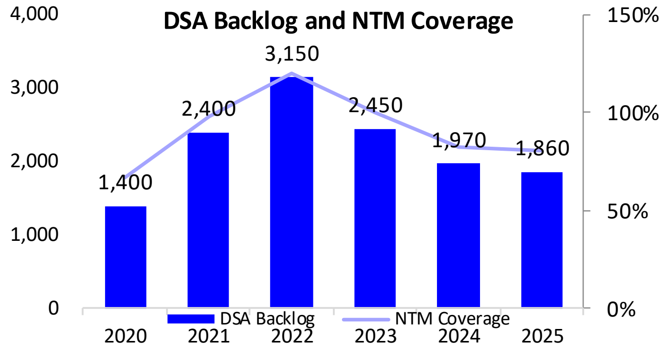
— What is Charles River's current revenue from NAMs what does the company view as the TAM for the offering?

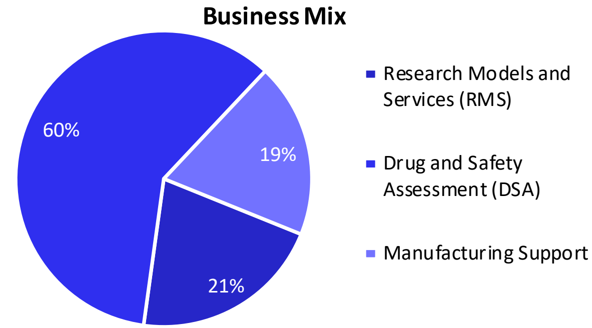
Justin Bowers; justin.bowers@db.com; +1.917.517.9836

Charles River Laboratories (CRL) / Buy / $260 Price Target

Company Overview







Financial Trends

IS Items ($ millions)	2023	2024	2025	2026E	2027E
Total Revenue	4,129	4,050	4,015	3,846	3,889
yoy chg.	*3.9%*	*(1.9%)*	*(0.9%)*	*(4.2%)*	*1.1%*
EBITDA	1,023	1,006	977	984	1,080
yoy chg.	*4.1%*	*(1.7%)*	*(2.9%)*	*0.7%*	*9.8%*
EBIT	839	806	794	809	894
yoy chg.	*(0.4%)*	*(3.9%)*	*(1.5%)*	*1.8%*	*10.5%*
EPS	$10.65	$10.32	$10.29	$11.00	$12.50
yoy chg.	*(4.3%)*	*(3.1%)*	*(0.3%)*	*7.0%*	*13.6%*

BS Items ($ millions)	2023	2024	2025	2026E	2027E
Cash	277	195	214	506	1,184
Total Debt	2,647	2,240	2,136	2,418	2,418
Net Debt	2,370	2,046	1,923	1,912	1,234
CF Items ($ millions)					
CFO	684	735	738	606	844
Free Cash Flow	365	502	498	399	678
Share Repurchases	(24)	(119)	(361)	(208)	0
Leverage Ratios					
Net Debt/EBITDA	2.3x	2.0x	2.0x	1.9x	1.1x

Valuation & Risks

Valuation

Price	$180.56
Market Cap ($M)	8,909
DB Rating	Buy
Valuation Methodology	13.5x our 2027 EBITDA
Price Target	$260.00

Risks

Downside risks to our price target include market pull-back, which would weaken biotech IPO incentives and thus VC funding, and challenges in integrating various acquisitions CRL has made.

Valuation Methodology

Our price target for CRL shares is 13.5x our 2027 EBITDA estimate, in line with the stock's 5-year historical average due to improving end-market conditions and sector multiples.

Source: Deutsche Bank, Company Filings, Bloomberg Finance LP
Note: Price & Mkt Cap as of 07/09/2026

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

CON: Top Questions
Economic data; market expansion; and new markets

- What is the relative market expansion Concentra is hoping to drive between organic growth / M&A for Onsite Health strategy; the company cited a $15 – 20b TAM only partially penetrated for onsite health clinics?

- Referenced that economic indicators point to things beginning to accelerate with net job gains for 2 months in a row, is there a way to think about a lag between improved economic data and how it translates to Concentra's outlook?

- Referenced onshoring as a potential tailwind to the industry, what is Concentra sizing the additional onshoring opportunity as and how is it thinking when the bulk of the build outs occur?

- How should investors be modeling the different fee schedule step ups yet to occur in 2026?

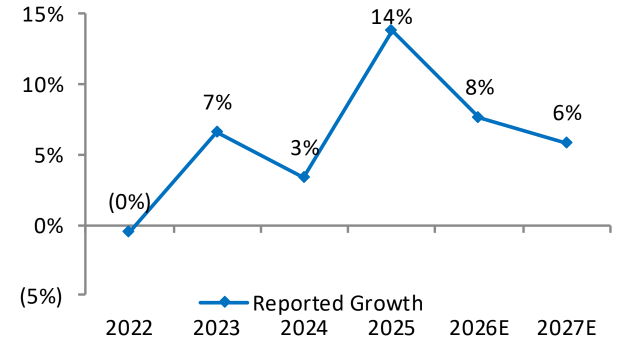
- What is the shortest / longest timeline for when the rate decision could occur and drive entrance into the NY state market?

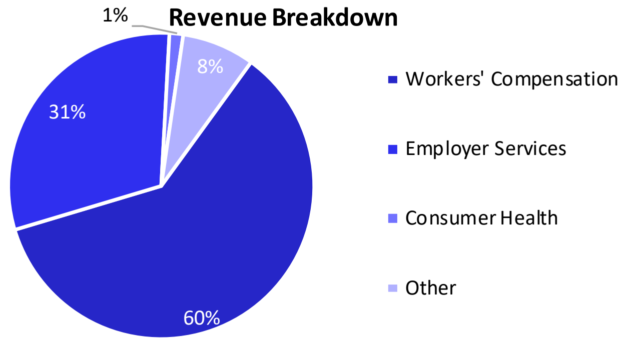
- Now expected 8 – 10 de Novos in 2026, spread across several new states. What level of economic data is required for the company to pull the trigger on entering a new market?

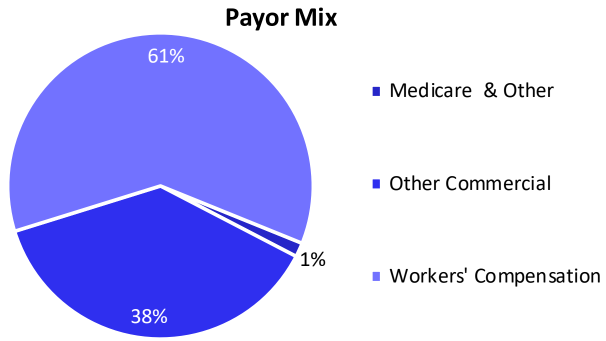
Justin Bowers; justin.bowers@db.com; +1.917.517.9836

Concentra (CON) / Buy / $32 Price Target

Company Overview







Financial Trends

IS Items ($ millions)	2023	2024	2025	2026E	2027E
Total Revenue	1,838	1,900	2,163	2,329	2,465
yoy chg.	*6.6%*	*3.4%*	*13.9%*	*7.7%*	*5.8%*
EBITDA	361	376	432	472	507
yoy chg.	*8.1%*	*4.1%*	*14.8%*	*9.2%*	*7.6%*
EBIT	288	307	346	373	406
yoy chg.	*11.3%*	*6.7%*	*12.6%*	*7.9%*	*8.8%*
EPS	$1.73	$1.46	$1.36	$1.51	$1.72
yoy chg.	*7.7%*	*(15.1%)*	*(7.1%)*	*11.1%*	*13.9%*

BS Items ($ millions)	2023	2024	2025	2026E	2027E
Cash	31	183	80	155	249
Total Debt	475	1,479	1,574	1,522	1,462
Net Debt	443	1,296	1,494	1,367	1,213
CF Items ($ millions)					
CFO	234	275	279	285	316
Free Cash Flow	169	210	197	214	243
Leverage Ratios					
Net Debt/EBITDA	1.2x	3.4x	3.5x	2.9x	2.4x

Valuation & Risks

Valuation

Price	$31.40
Market Cap ($M)	$4,018
DB Rating	Buy
Valuation Methodology	11.7x 2026 EV/EBITDA
Price Target	$32.00

Source: Deutsche Bank, Company Filings, Bloomberg Finance LP
Note: Price & Mkt Cap as of 07/09/2026

Risks

Downside risks: weakening macroeconomic environment, state reimbursement cuts, contract losses, margin contraction, litigation, geographic concentration.

Valuation Methodology

Our price target for CON stock reflects a multiple of 11x 2026 EV/EBITDA marking a double-digit discount to peers, which we believe is appropriate as a new standalone public entity. However, we do believe the stock could close the discount with consistent execution.

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

FTRE: Top Questions

2H26 outlook; improved EBP bookings / performance; cost savings; and AI opportunity

— How should investors be thinking about any potential QoQ acceleration in 2H26 as the increase in awards / win-rates begins to reflect on the top line? Is there going to be a bigger bolus represented in 4Q?

— What sort of QoQ ramp on EBITDA should we be modeling when considering the mix of pass-throughs and the abnormally large percentage of EBITDA (relative to guidance) that occurred in year 1?

— Large pharma remains stable while EBP continues to recover as funding improves, on EBP how has activity / bookings from this cohort trended from 4Q25 to 1Q26 and now into 2Q26?

— Book-to-bill improvement partially driven by an intensified focus on EBP, what sort of magnitude of improvement was seen from EBP authorizations? What was the increase either in dollars or percentage on new to Fortrea EBP?

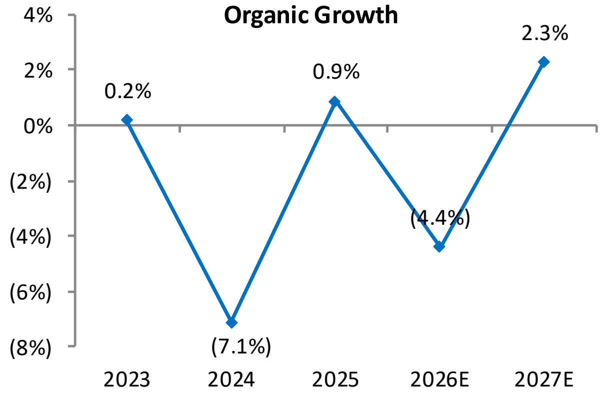
— For the cost savings initiatives, is there any way to quantify the benefit to margins in terms of number of dollars of gross or net savings achieved and the pace of realization of the benefits?

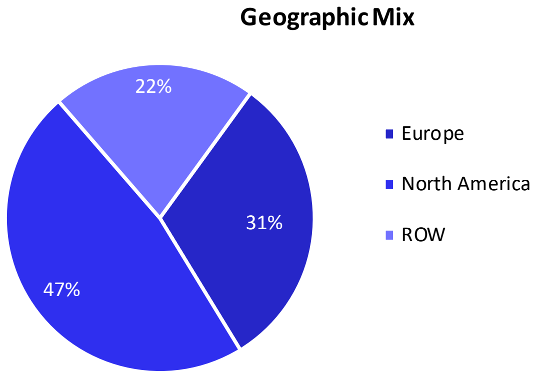
— Highlighted the FIT initiative and its initial reception, what was the total opportunity sized at and how do we think about ROIC on AI-related investments and the timeline to achieve it relative to other projects Fortrea has undertaken?

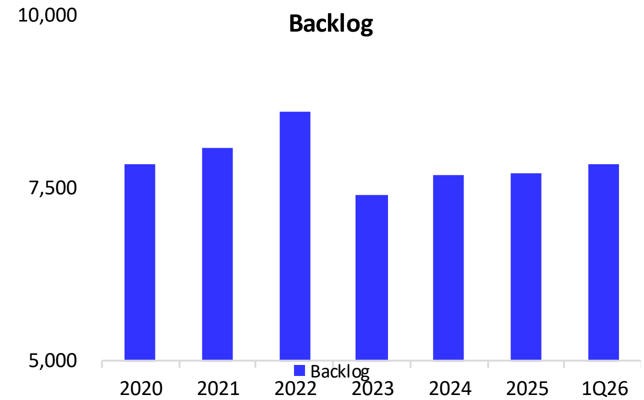
Justin Bowers; justin.bowers@db.com; +1.917.517.9836

Fortrea (FTRE) / Hold / $18 Price Target

Company Overview







Financial Trends

IS Items ($ millions)	2023	2024	2025	2026E	2027E
Total Revenue	2,843	2,696	2,723	2,609	2,668
yoy chg.	*(8.2%)*	*(5.1%)*	*1.0%*	*(4.2%)*	*2.3%*
EBITDA	246	205	190	206	220
yoy chg.	*(39.3%)*	*(16.6%)*	*(7.3%)*	*8.6%*	*6.4%*
EBIT	216	166	141	174	198
yoy chg.	*(35.1%)*	*(23.2%)*	*(15.3%)*	*23.7%*	*13.5%*
EPS	$1.26	$0.35	$0.43	$0.83	$0.99
yoy chg.	*(62.8%)*	*(72.3%)*	*24.3%*	*91.4%*	*19.1%*

BS Items ($ millions)	2023	2024	2025	2026E	2027E
Cash	109	119	175	214	223
Total Debt	1,592	1,125	1,062	979	894
Net Debt	1,483	1,006	887	764	671
CF Items ($ millions)					
CFO	168	263	114	149	126
Free Cash Flow	128	237	88	117	93
Share Repurchases	0	0	0	0	0
Leverage Ratios					
Net Debt/EBITDA	6.0x	4.9x	4.7x	3.7x	3.1x

Valuation & Risks

Valuation

Price	$17.67
Market Cap ($M)	1,672
DB Rating	Hold
Valuation Methodology	11x 2027 EBITDA
Price Target	$18.00

Risks

Upside risks: outsized margin expansion; faster-than-expected TSA exits; market share gains; robust biotech funding environment. Downside risks: slow or limited margin expansion; post-spin execution issues; market share losses; quality or regulatory issues; subdued biotech fundraising environment.

Valuation Methodology

Our price target for FTRE stock reflects a multiple of 11x 2027 EV/EBITDA. Our target multiple is a discount to the CRO peer group median, with greater margin expansion and EBITDA growth prospects offset by greater financial risk from elevated leverage and revenue growth headwinds this year.

Source: Deutsche Bank, Company Filings, Bloomberg Finance LP
Note: Price & Mkt Cap as of 07/09/2026

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

MRVI: Top Questions

Demand metrics; new product uptake; EBITDA margin expansion; and AI-based offerings

— Conversion rate improved in the quarter and is driving faster-than-expected discovery orders while trial activity by phase was stable in 1Q26, YTD how did these metrics trend from January to now? How are they trending QoQ so far?

— Tri-link forecast raised to +high-teens YoY, with the forecast raised due to strength from GMP consumables and a return to growth in discovery, what does updated guidance require on a YoY basis from GMP and discovery, respectively?

— Updated EBITDA guidance implies ~15% margin in 2026, how should investors be thinking about incrementals by product (GMP vs. discovery vs. Cygnus) over the course of 2026 and the respective contributions to margin expansion?

— ModTail performance > initial expectations with 70 customers and early demand for GMP-quality products, what was the revenue contribution from ModTail in the quarter and how has the initial strength informed management's expectations for the product in 2026?

— AI-enabled mRNA Builder gaining traction as well, what's the initial revenue contribution from AI solutions for Maravai and what is expected in 2026? What proportion of customer studies are leveraging AI for trial / target design?

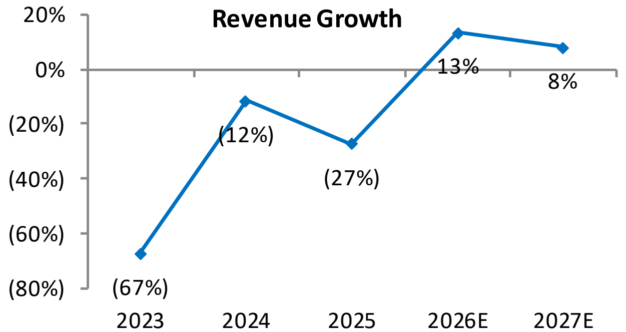
— While Cygnus maintains its 100% attachment rate for all 29 FDA / EMA-approved C&GTs, what does Maravai anticipate in terms of pipeline opportunities for the kits in 2026 /2027?

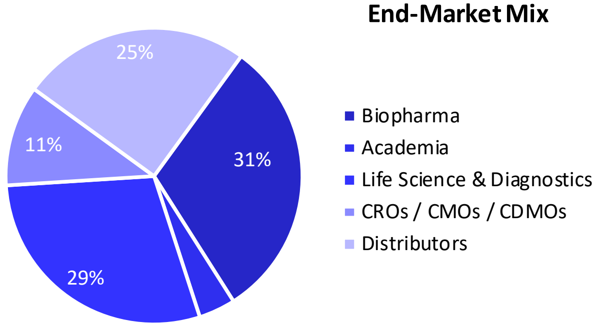
Justin Bowers; justin.bowers@db.com; +1.917.517.9836

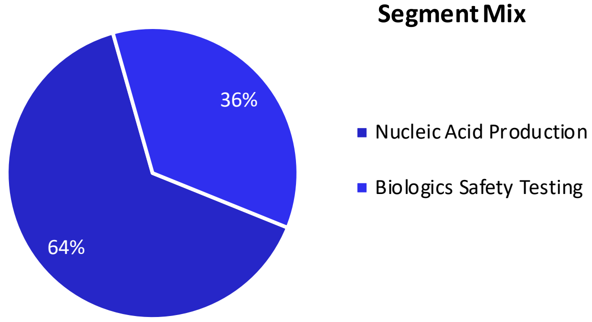
Maravai One Pager

Maravi Life Sciences (MRVI) / Buy / $6.00 Price Target

Company Overview







Financial Trends

IS Items ($ millions)	2023	2024	2025	2026E	2027E
Total Revenue	289	255	186	211	228
yoy chg.	*(67.3%)*	*(11.6%)*	*(27.3%)*	*13.4%*	*8.0%*
EBITDA	65	32	(31)	30	42
yoy chg.	*(89.8%)*	*(50.3%)*	*(196.2%)*	*195.2%*	*41.2%*
EBIT	17	(15)	(82)	(16)	(4)
yoy chg.	*(97.2%)*	*(188.4%)*	*457.2%*	*81.0%*	*71.4%*
EPS	$0.03	($0.11)	($0.29)	($0.09)	($0.05)
yoy chg.	*(98.5%)*	*(524.2%)*	*157.2%*	*69.6%*	*38.7%*

BS Items ($ millions)	2023	2024	2025	2026E	2027E
Cash	575	322	217	228	274
Total Debt	524	296	292	241	241
Net Debt	(51)	(26)	75	13	(33)
CF Items ($ millions)					
CFO	126	7	(58)	74	55
Free Cash Flow	61	(22)	(71)	66	47
Share Repurchases	0	0	0	0	0
Leverage Ratios					
Net Debt/EBITDA	(0.8x)	(0.8x)	(2.4x)	0.5x	(0.8x)

Valuation & Risks

Valuation

Price	$5.97
Market Cap ($M)	1,541
DB Rating	Buy
Valuation Methodology	6.5x 2027 EV / Sales
Price Target	$6.00

Source: Deutsche Bank, Company Filings, Bloomberg Finance LP
Note: Price & Mkt Cap as of 07/09/2026

Risks

Downside risk: C19 backfill issues; margin degradation; decreased or selective biopharma interest in mRNA technologies; intellectual property issues.

Valuation Methodology

Our target price for MRVI shares represents a multiple of 6.5x 2027 EV / Sales estimate. This is a one turn discount to its historical trading range given the improving visibility into the inflection in revenue growth and operation improvements as evidenced by the company's margin turnaround and positive FCF generation.

Justin Bowers; justin.bowers@db.com; +1.917.517.9836

Important Disclosures

*Other information available upon request

Analyst Certification

The views expressed in this report accurately reflect the personal views of the undersigned lead analyst about the subject issuers and the securities of those issuers. In addition, the undersigned lead analyst has not and will not receive any compensation for providing a specific recommendation or view in this report. **Justin Bowers**

Company Rating and Dispersion Key: The table provides a snapshot of Deutsche Bank's company research rating distribution across our covered companies. We also present the percentage of companies where Deutsche Bank has provided Investment Banking Services in the past 12 months and/or MIFID Investment and Ancillary services, in the past 12 months. Please see the key and definition of our rating below.
Note - percentages are rounded so may not total 100%.
Covered: The overall rating distribution across all companies under coverage with a rating.
w/Banking relation: Percentage of companies under coverage with a rating within each of the "buy", "hold" and "sell" categories for which Deutsche Bank has provided Investment Banking Services within the previous 12 months.
w/MiFID services: Percentage of companies under coverage with a rating within each of the "buy", "hold" and "sell" categories for which Deutsche Bank has provided MIFID Investment and Ancillary services within the previous 12 months.
Buy/Hold/Sell Percentages: These percentages reflect the proportion of companies within each category that have been assigned the corresponding rating, based on our 12-month view of Total Shareholder Return (TSR).

Rating definitions:
Buy: Based on a current 12-month view of TSR, we recommend that investors buy the stock.
Sell: Based on a current 12-month view of TSR, we recommend that investors sell the stock.
Hold: We take a neutral view on the stock 12-months out and, based on this time horizon, do not recommend either a Buy or Sell.
TSR: Total Shareholder Return. Percentage change in share price from current price to projected target price plus projected dividend yield.
Newly issued research recommendations and target prices supersede previously published research.

Company Rating Dispersion and Banking Relationships

Company rating dispersion and banking relationships

DBSI Companies under Coverage	Buy	Hold	Sell
Covered	57%	43%	0%
w/ Banking relationship	42%	35%	0%
w/ MiFID services	62%	51%	67%

Additional Information

The information and opinions in this report were prepared by Deutsche Bank AG or one of its affiliates (collectively "Deutsche Bank"). Though the information herein is believed to be reliable and has been obtained from public sources believed to be reliable, Deutsche Bank makes no representation as to its accuracy or completeness. Hyperlinks to third-party websites in this report are provided for reader convenience only. Deutsche Bank neither endorses the content nor is responsible for the accuracy or security controls of those websites.

If you use the services of Deutsche Bank in connection with a purchase or sale of a security that is discussed in this report, or is included or discussed in another communication (oral or written) from a Deutsche Bank analyst, Deutsche Bank may act as principal for its own account or as agent for another person.

Deutsche Bank may consider this report in deciding to trade as principal. It may also engage in transactions, for its own account or with customers, in a manner inconsistent with the views taken in this research report. Others within Deutsche Bank, including strategists, sales staff and other analysts, may take views that are inconsistent with those taken in this research report. Deutsche Bank issues a variety of research products, including fundamental analysis, equity-linked analysis, quantitative analysis and trade ideas. Recommendations contained in one type of communication may differ from recommendations contained in others, whether as a result of differing time horizons, methodologies, perspectives or otherwise. Deutsche Bank and/or its affiliates may also be holding debt or equity securities of the issuers it writes on. Analysts are paid in part based on the profitability of Deutsche Bank AG and its affiliates, which includes investment banking, trading and principal trading revenues.

Opinions, estimates and projections constitute the current judgment of the author as of the date of this report. They do not necessarily reflect the opinions of Deutsche Bank and are subject to change without notice. Deutsche Bank provides liquidity for buyers and sellers of securities issued by the companies it covers. Deutsche Bank research analysts sometimes have shorter-term trade ideas that may be inconsistent with Deutsche Bank's existing longer-term ratings. Some trade ideas for equities are listed as Catalyst Calls ideas on the Research Website (https://research.db.com/Research/), and can be found on the general coverage list and also on the covered company's page. A Catalyst Call idea represents a high-conviction belief by an analyst that a stock will outperform or underperform the market and/or a specified sector over a time frame of no less than two weeks and no more than three months. In addition to Catalyst Calls ideas, analysts may occasionally discuss with our clients, and with Deutsche Bank salespersons and traders, trading strategies or ideas that reference catalysts or events that may have a near-term or medium-term impact on the market price of the securities discussed in this report, which impact may be directionally counter to the analysts' current 12-month view of total return or investment return as described herein. Deutsche Bank has no obligation to update, modify or amend this report or to otherwise notify a recipient thereof if an opinion, forecast or estimate changes or becomes inaccurate. Coverage and the frequency of changes in market conditions and in both general and company-specific economic prospects make it difficult to update research at defined intervals. Updates are at the sole discretion of the coverage analyst or of the Research Department Management, and the majority of reports are published at irregular intervals. This report is provided for informational purposes only and does not take into account the particular investment objectives, financial situations, or needs of individual clients. It is not an offer or a solicitation of an offer to buy or sell any financial instruments or to participate in any particular trading strategy. Target prices are inherently imprecise and a product of the analyst's judgment. The financial instruments discussed in this report may not be suitable for all investors, and investors must make their own informed investment decisions. Prices and availability of financial instruments are subject to change without notice, and investment transactions can lead to losses as a result of price fluctuations and other factors. If a financial instrument is denominated in a currency other than an investor's currency, a change in exchange rates may adversely affect the investment. Past performance is not necessarily indicative of future results. Performance calculations exclude transaction costs, unless otherwise indicated. Unless otherwise indicated, prices are current as of the end of the previous trading session and are sourced from local exchanges via Reuters, Bloomberg and other vendors. Data is also sourced from Deutsche Bank, subject companies, and other parties. Artificial intelligence tools may be used in the preparation of this material, including but not limited to assist in fact-finding, data analysis, pattern recognition, content drafting and editorial corrections pertaining to research material.

The Deutsche Bank Research Department is independent of other business divisions of the Bank. Details regarding our organizational arrangements and information barriers we have to prevent and avoid conflicts of interest with respect to our research are available on our website (https://research.db.com/Research/) under Disclaimer.

Macroeconomic fluctuations often account for most of the risks associated with exposures to instruments that promise to pay fixed or variable interest rates. For an investor who is long fixed-rate instruments (thus receiving these cash flows), increases in interest rates naturally lift the discount factors applied to the expected cash flows and thus cause a loss. The longer the maturity of a certain cash flow and the higher the move in the discount factor, the higher will be the loss. Upside surprises in inflation, fiscal funding needs, and FX depreciation rates are among the most common adverse macroeconomic shocks to receivers. But counterparty exposure, issuer creditworthiness, client segmentation, regulation (including changes in assets holding limits for different types of investors), changes in tax policies, currency convertibility (which may constrain currency conversion, repatriation of profits and/or liquidation of positions), and settlement issues related to local clearing houses are also important risk factors. The sensitivity of fixed-income instruments to macroeconomic shocks may be mitigated by indexing the contracted cash flows to inflation, to FX depreciation, or to specified interest rates – these are common in emerging markets. The index fixings may – by construction – lag or mis-measure the actual move in the underlying variables they are intended to track. The choice of the proper fixing (or metric) is particularly important in swaps markets, where floating coupon rates (i.e., coupons indexed to a typically short-dated interest rate reference index) are exchanged for fixed coupons. Funding in a currency that differs from the currency in which coupons are denominated carries FX risk. Options on swaps (swaptions) the risks typical to options in addition to the risks related to rates movements.

Derivative transactions involve numerous risks including market, counterparty default and illiquidity risk. The appropriateness of these products for use by investors depends on the investors' own circumstances, including their tax position, their regulatory environment and the nature of their other assets and liabilities; as such, investors should take expert legal and financial advice before entering into any transaction similar to or inspired by the contents of this publication. The risk of loss in futures trading and options, foreign or domestic, can be substantial. As a result of the high degree of leverage obtainable in futures and options trading, losses may be incurred that are greater than the amount of funds initially deposited – up to theoretically unlimited losses. Trading in options involves risk and is not suitable for all investors. Prior to buying or selling an option, investors must review the "Characteristics and Risks of Standardized Options", at https://www.theocc.com/company-information/documents-and-archives/options-disclosure-document. If you are unable to access the website, please contact your Deutsche Bank representative for a copy of this important document.

Participants in foreign exchange transactions may incur risks arising from several factors, including the following: (i) exchange rates can be volatile and are subject to large fluctuations; (ii) the value of currencies may be affected by numerous market factors, including world and national economic, political and regulatory events, events in equity and debt markets and changes in interest rates; and (iii) currencies may be subject to devaluation or government-imposed exchange controls, which could affect the value of the currency. Investors in securities such as ADRs, whose values are affected by the currency of an underlying security, effectively assume currency risk.

Unless governing law provides otherwise, all transactions should be executed through the Deutsche Bank entity in the investor's home jurisdiction. Aside from within this report, important conflict disclosures can also be found at https://research.db.com/Research/ on each company's research page or under the 'Disclosures' tab . Investors are strongly encouraged to review this information before investing.

Deutsche Bank (which includes Deutsche Bank AG, its branches and affiliated companies) is not acting as a financial adviser, consultant or fiduciary to you or any of your agents (collectively, "You" or "Your") with respect to any information provided in this report. Deutsche Bank does not provide investment, legal, tax or accounting advice, Deutsche Bank is not acting as your impartial adviser, and does not express any opinion or recommendation whatsoever as to any strategies, products or any other information presented in the materials. Information contained herein is being provided solely on the basis that the recipient will make an independent assessment of the merits of any investment decision, and it does not constitute a recommendation of, or express an opinion on, any product or service or any trading strategy.

The information presented is general in nature and is not directed to retirement accounts or any specific person or account type, and is therefore provided to You on the express basis that it is not advice, and You may not rely upon it in making Your decision. The information we provide is being directed only to persons we believe to be financially sophisticated, who are capable of evaluating investment risks independently, both in general and with regard to particular transactions and investment strategies, and who understand that Deutsche Bank has financial interests in the offering of its products and services. If this is not the case, or if You are an IRA or other retail investor receiving this directly from us, we ask that you inform us immediately.

In July 2018, Deutsche Bank revised its rating system for short term ideas whereby the branding has been changed to Catalyst Calls ("CC") from SOLAR ideas; the rating categories for Catalyst Calls originated in the Americas region have been made consistent with the categories used by Analysts globally; and the effective time period for CCs has been reduced from a maximum of 180 days to 90 days.

United States: Approved and/or distributed by Deutsche Bank Securities Incorporated, a member of FINRA and SIPC. Analysts located outside of the United States are employed by non-US affiliates and are not registered/qualified as research analysts with FINRA.

European Economic Area (exc. United Kingdom): Approved and/or distributed by Deutsche Bank AG, a joint stock corporation with limited liability incorporated in the Federal Republic of Germany with its principal office in Frankfurt am Main. Deutsche Bank AG is authorized under German Banking Law and is subject to supervision by the European Central Bank and by BaFin, Germany's Federal Financial Supervisory Authority.

United Kingdom: Approved and/or distributed by Deutsche Bank AG acting through its London Branch at 21 Moorfields, London EC2Y 9DB. Deutsche Bank AG in the United Kingdom is authorised by the Prudential Regulation Authority and is subject to limited regulation by the Prudential Regulation Authority and Financial Conduct Authority. Details about the extent of our authorisation and regulation are available on request.

Hong Kong SAR: Distributed by Deutsche Bank AG, Hong Kong Branch except for any research content relating to futures contracts within the meaning of the Hong Kong Securities and Futures Ordinance Cap. 571. Research reports on such futures contracts are not intended for access by persons who are located, incorporated, constituted or resident in Hong Kong. The author(s) of a research report may not be licensed to carry on regulated activities in Hong Kong and, if not licensed, do not hold themselves out as being able to do so. The provisions set out above in the "Additional Information" section shall apply to the fullest extent permissible by local laws and regulations, including without limitation the Code of Conduct for Persons Licensed or Registered with the Securities and Futures Commission. This report is intended for distribution only to "professional investors" as defined in Part 1 of Schedule 1 of the SFO. This document must not be acted or relied on by persons who are not professional investors. Any investment or investment activity to which this document relates is only available to professional investors and will be engaged only with professional investors.

India: Prepared by Deutsche Equities India Private Limited (DEIPL) having CIN: U65990MH2002PTC137431 and registered office at 14th Floor, The Capital, C-70, G Block, Bandra Kurla Complex, Mumbai (India) 400051. Tel: + 91 22 7180 4444. It is registered by the Securities and Exchange Board of India (SEBI) as a Stock broker bearing registration no.: INZ000252437; Merchant Banker bearing SEBI Registration no.: INM000010833 and Research Analyst bearing SEBI Registration no.: INH000001741. DEIPL's Compliance / Grievance officer is Ms. Rashmi Poddar (Tel: +91 22 7180 4929, email ID: complaints.deipl@db.com). Registration granted by SEBI and certification from NISM in no way guarantee performance of DEIPL or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. DEIPL may have received administrative warnings from the SEBI for breaches of Indian regulations. Deutsche Bank and/or its affiliate(s) may have debt holdings or positions in the subject company. With regard to information on associates, please refer to the "Shareholdings" section in the Annual Report at: https://www.db.com/ir/en/annual-reports.htm . For latest India research audit report, refer https://country.db.com/india/deutsche-equities-india/index?language_id=1.

Japan: Approved and/or distributed by Deutsche Securities Inc.(DSI). Registration number - Registered as a financial instruments dealer by the Head of the Kanto Local Finance Bureau (Kinsho) No. 117. Member of associations: JSDA, Type II Financial Instruments Firms Association and The Financial Futures Association of Japan. Commissions and risks involved in stock transactions - for stock transactions, we charge stock commissions and consumption tax by multiplying the transaction amount by the commission rate agreed with each customer. Stock transactions can lead to losses as a result of share price fluctuations and other factors. Transactions in foreign stocks can lead to additional losses stemming from foreign exchange fluctuations. We may also charge commissions and fees for certain categories of investment advice, products and services. Recommended investment strategies, products and services carry the risk of losses to principal and other losses as a result of changes in market and/or economic trends, and/or fluctuations in market value. Before deciding on the purchase of financial products and/or services, customers should carefully read the relevant disclosures, prospectuses and other documentation. "Moody's", "Standard & Poor's", and "Fitch" mentioned in this report are not registered credit rating agencies in Japan unless Japan or "Nippon" is specifically designated in the name of the entity. Reports on Japanese listed companies not written by analysts of DSI are written by Deutsche Bank Group's analysts with the coverage companies specified by DSI. Some of the foreign securities stated on this report are not disclosed according to the Financial Instruments and Exchange Law of Japan. Target prices set by Deutsche Bank's equity analysts are based on a 12-month forecast period.

Korea: Distributed by Deutsche Securities Korea Co.

South Africa: Deutsche Bank AG Johannesburg is incorporated in the Federal Republic of Germany (Branch Register Number in South Africa: 1998/003298/10).

Singapore: This report is issued by Deutsche Bank AG, Singapore Branch (One Raffles Quay #18-00 South Tower Singapore 048583, +65 6423 8001), which may be contacted in respect of any matters arising from, or in connection with, this report. Where this report is issued or promulgated by Deutsche Bank in Singapore to a person who is not an accredited investor, expert investor or institutional investor (as defined in the applicable Singapore laws and regulations), they accept legal responsibility to such person for its contents.

Taiwan: Information on securities/investments that trade in Taiwan is for your reference only. Readers should independently evaluate investment risks and are solely responsible for their investment decisions. Deutsche Bank research may not be distributed to the Taiwan public media or quoted or used by the Taiwan public media without written consent. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation to trade in such securities/instruments.

Qatar: Deutsche Bank AG in the Qatar Financial Centre (registered no. 00032) is regulated by the Qatar Financial Centre Regulatory Authority. Deutsche Bank AG - QFC Branch may undertake only the financial services activities that fall within the scope of its existing QFCRA license. Its principal place of business in the QFC: Qatar Financial Centre, Tower, West Bay, Level 5, PO Box 14928, Doha, Qatar. This information has been distributed by Deutsche Bank AG. Related financial products or services are only available only to Business Customers, as defined by the Qatar Financial Centre Regulatory Authority.

Russia: The information, interpretation and opinions submitted herein are not in the context of, and do not constitute, any appraisal or evaluation activity requiring a license in the Russian Federation.

Kingdom of Saudi Arabia: Deutsche Securities Saudi Arabia (DSSA) is a closed joint stock company authorized by the Capital Market Authority of the Kingdom of Saudi Arabia with a license number (No. 37-07073) to conduct the following business activities: Dealing, Arranging, Advising, and Custody activities. . DSSA registered office is at Faisaliah Tower, 17th floor, King Fahad Road - Al Olaya District Riyadh, Kingdom of Saudi Arabia P.O. Box 301806.

United Arab Emirates: Deutsche Bank AG in the Dubai International Financial Centre (registered no. 00045) is regulated by the Dubai Financial Services Authority. Deutsche Bank AG - DIFC Branch may only undertake the financial services activities that fall within the scope of its existing DFSA license. Principal place of business in the DIFC: Dubai International Financial Centre, The Gate Village, Building 5, PO Box 504902, Dubai, U.A.E. This information has been distributed by Deutsche Bank AG. Related financial products or services are available only to Professional Clients, as defined by the Dubai Financial Services Authority.

Australia and New Zealand: This research is intended only for "wholesale clients" within the meaning of the Australian Corporations Act and New Zealand Financial Advisors Act, respectively. Please refer to Australian specific research disclosures and related information at https://www.dbresearch.com/PROD/RPS_EN-PROD/PROD0000000000521304.xhtml . Where research refers to any particular financial product recipients of the research should consider any product disclosure statement, prospectus or other applicable disclosure document before making any decision about whether to acquire the product. In preparing this report, the primary analyst or an individual who assisted in the preparation of this report has likely been in contact with the company that is the subject of this research for confirmation/clarification of data, facts, statements, permission to use company-sourced material in the report, and/or site-visit attendance. Without prior approval from Research Management, analysts may not accept from current or potential Banking clients the costs of travel, accommodations, or other expenses incurred by analysts attending site visits, conferences, social events, and the like. Similarly, without prior approval from Research Management and Anti-Bribery and Corruption ("ABC") team, analysts may not accept perks or other items of value for their personal use from issuers they cover.

Additional information relative to securities, other financial products or issuers discussed in this report is available upon request. This report may not be reproduced, distributed or published without Deutsche Bank's prior written consent.

Backtested, hypothetical or simulated performance results have inherent limitations. Unlike an actual performance record based on trading actual client portfolios, simulated results are achieved by means of the retroactive application of a backtested model itself designed with the benefit of hindsight. Taking into account historical events the backtesting of performance also differs from actual account performance because an actual investment strategy may be adjusted any time, for any reason, including a response to material, economic or market factors. The backtested performance includes hypothetical results that do not reflect the reinvestment of dividends and other earnings or the deduction of advisory fees, brokerage or other commissions, and any other expenses that a client would have paid or actually paid. No representation is made that any trading strategy or account will or is likely to achieve profits or losses similar to those shown. Alternative modeling techniques or assumptions might produce significantly different results and prove to be more appropriate. Past hypothetical backtest results are neither an indicator nor guarantee of future returns. Actual results will vary, perhaps materially, from the analysis.

The method for computing individual E,S,G and composite ESG scores set forth herein is a novel method developed by the Research department within Deutsche Bank AG, computed using a systematic approach without human intervention. Different data providers, market sectors and geographies approach ESG analysis and incorporate the findings in a variety of ways. As such, the ESG scores referred to herein may differ from equivalent ratings developed and implemented by other ESG data providers in the market and may also differ from equivalent ratings developed and implemented by other divisions within the Deutsche Bank Group. Such ESG scores also differ from other ratings and rankings that have historically been applied in research reports published by Deutsche Bank AG. Further, such ESG scores do not represent a formal or official view of Deutsche Bank AG.

It should be noted that the decision to incorporate ESG factors into any investment strategy may inhibit the ability to participate in certain investment opportunities that otherwise would be consistent with your investment objective and other principal investment strategies. The returns on a portfolio consisting primarily of sustainable investments may be lower or higher than portfolios where ESG factors, exclusions, or other sustainability issues are not considered, and the investment opportunities available to such portfolios may differ. Companies may not necessarily meet high performance standards on all aspects of ESG or sustainable investing issues; there is also no guarantee that any company will meet expectations in connection with corporate responsibility, sustainability, and/or impact performance.